

Review

Voluntary Carbon Markets (VCMs) in a nutshell: A systematic review based on the empirical evidence from across the globe

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HIGHLIGHTS

- A critical appraisal of the key findings reveals an urgent requirement for a policy framework for VCMs irrespective of the climate zone the study belongs to.
- A focus on computing the economic aspects of carbon projects (through ROI) is noted.
- A substantial difference in the nature of studies conducted in the Global North vis-à-vis the Global South is observed.
- An inclination towards agroforestry projects as opposed to the REDD mechanism is seen concerning optimal land use.
- Co-benefits of VCM projects play an important role in the implementation of the projects as well as in the sale of the carbon credits so generated.

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ABSTRACT

Voluntary Carbon Markets (VCMs) have attracted various stakeholders to participate in their value chain as they allow carbon credits to be traded. This study systematically reviews the academic literature pertinent to VCMs combined with an analysis of the selected studies on certain bases such as classification based on the climate zones & regions, Global North- Global South perspective, and land use perspective. Further, it also explores the broad themes, the research timeline, and the empirical methods used in the extant literature. Additionally, the study digs into the gendered dimension in certain VCM projects. To fulfill these purposes, sixty-nine studies based on empirical evidence have been chosen from the vast literature. A critical appraisal of the key findings reveals an urgent requirement for a policy framework for VCMs irrespective of the climate zone the study belongs to. A focus on the financial and socio-economic aspects of carbon projects is noted and a substantial difference in the nature of studies conducted in Global North/Global South is observed. Effective land use regarding superior alternatives is cited in informing policymakers and project developers of financially, environmentally, and socially feasible uses. The gender stereotypes revolving around certain VCM-based projects need to be addressed. Drawing upon the critical findings of this study, possible future work in the domain of carbon pricing in VCM, investigating efficient land use to combat climate change, “Women and ESG”, and the unified role of compliance and voluntary markets in mitigating climate change is suggested.

1. Introduction

Voluntary Carbon Markets (VCMs) have gained significant attention in recent years due to the increasing need to adopt climate change measures. Broadly, VCM is one of the two types of Carbon Markets, the other being Compliance Markets. Carbon markets are trading systems in which carbon credits (wherein one tonne of CO₂ or other greenhouse gases emission avoided or sequestered equals one credit) are sold and bought by companies and individuals. Such a market mechanism is considered important to control the amount of greenhouse gases in the

atmosphere to tackle climate change (UNEP, 2025; Pierce and Strong, 2023). While compliance markets are mandated by national/international regulations to achieve emission targets, VCMs are driven by voluntary participation of private entities to meet their emission goals. VCMs trace their evolution back to 1997 when the adoption of the Kyoto Protocol led to the establishment of global compliance carbon market mechanisms for the first time (Spilker and Nugent, 2022). Initially subdued by the growth of compliance markets till the late 2000s, many companies and governments gradually began participating in parallel voluntary mechanisms after gaining confidence in these markets

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(Benessaiah, 2012).

Organizations’ interest in VCM has nearly tripled, defying the economic effects of COVID-19 as well. These markets have the potential to be a major force behind mitigation efforts if properly managed, especially in developing economies that provide the majority of voluntary carbon offsets (Streck, 2021). Remarkably, developing countries are also at a higher economic and environmental risk due to green-house gas emissions (Shaari et al., 2024; Sutikno et al., 2024), and their increasing emission responsibility requires greater attention (Meng et al., 2023). Carbon offsets are based on carbon credits generated through carbon offset projects linked to VCMs that the corporations are making claims regarding their past and future climate change mitigation initiatives. The intended use of these carbon credits can be (1) generation of credits for offsetting or non-offsetting uses, or (2) to fulfill the net-zero targets or meet carbon-neutral goals (Trouwloon et al., 2023).

Various sectors such as agriculture, forestry, and energy can be targeted to establish VCM projects through efficient land use and choice of sustainable alternatives. By implementing environmentally friendly farming methods like zero tillage, laser-assisted precision land leveling, direct rice seeding, intercropping, applying biochar, using solar energy, and more effective management of irrigation water, soil nutrients, livestock feed, and manure, the agriculture sector can move from being a net emitter of greenhouse gases to a net absorber of them (Khurana et al., 2024). Additionally, larger investments in greener energy production projects can lead to faster returns on investment than other alternatives (Krach et al., 2024). Further, Community-based Forest Management projects linked to VCM are an economically viable option for sequestering carbon and promoting development through better livelihoods and infrastructure (Lupala et al., 2017). Remarkably, the other sectors participating in VCM are (Table 1) Transport, Households, Land Management, Waste Management, Industrial emissions, and carbon capture (University of California, 2025; Dev and Krishnamurthy, 2023).

However, as these markets show remarkable growth across sectors and throughout the globe, their governance has become increasingly complex and a pressing issue (Ahonen et al., 2022). To add to it, there have also been debates about how emission rights, which were intended to be ideal commodities, now fail to pass their intended markets with ease. Following several market failures, traders have to give their offsets

a narrative that goes beyond the simple storage of carbon in soil or trees to establish their value (Lock, 2024). Stories centred around carbon offset establish a close relationship between the product (carbon credits), its origin context, and the customer’s context in voluntary marketplaces. In this way, they manage carbon offset by navigating intricate relationships between value generation, value exchange, measuring techniques, and consumer demand (Brill, 2021). Further challenges for the VCM appear in the form of -

- Poorly worded labels, lack of clarity and uniformity in guidelines across certifying parties, absence of trust for certifiers (when they are not independent third-parties), and label proliferation from the demand side of the market.
- In contrast, the supply side raises concerns in the guise of farmers reluctance to alter their production methods to produce carbon credits with an undetermined value, reluctance of lending organizations to provide capital to producers who might require particular assets for the manufacturing processes used to generate carbon credits in the face of an uncertain market (Wongpiyabovorn et al., 2023).
- Difficulty in carbon capturing due to high investment costs and technological uncertainties. It has been seen that the large-scale deployment of carbon capture, utilization, and storage (CCUS) technologies faces significant challenges (Zhou et al., 2024).

Balancing and facilitating the arduous route, VCMs still stand strong with positive momentum as the Conference of Parties (COP) meetings of the United Nations discuss climate change annually and push towards efforts from the member nations to establish carbon markets and meet their Nationally Determined Commitments (Asadnabizadeh and Moe, 2024). The COP-29 meeting, which wrapped up in Baku in 2024, enumerated the much-desired international framework for VCMs. This credibility push combined with the ability to engage with diverse, impactful projects while addressing personal or organizational goals makes the VCM an exciting area to explore. Hence, the objective of this study is to investigate the relevant literature of VCMs and establish the academic contributions made in this field and critically appraise the findings to understand the current scenario and future possibilities in this domain.

1.1. Identifying the knowledge gap

A probe into the existing literature related to VCMs revealed a diverse yet scattered collection of research papers, book chapters, as well as conference proceedings articles. It was found that different aspects of VCMs across different geographies covering various themes have been explored. Several studies mentioned the existence of co-benefits as the primary motivation for establishing VCM-linked carbon projects (Teo, 2024; Paiva et al., 2014; Grace et al., 2010). Further, several studies highlighted that the economic viability of projects improved due to adopting sustainable management practices, especially in forestry projects (Ezquerro et al., 2024; Moreno et al., 2019). A few studies also assessed the total carbon stock capacity of various nature-based solutions and commented on their qualification as an existing successful carbon project or a potential future carbon project (Singh et al., 2024; Pinnschmidt et al., 2023). Governance issues were highlighted (Meitern, 2024; Sorge et al., 2022) as well as dual-financing sources were discussed and suggested for the profitability of the carbon projects (Li and Martino, 2024; Moxey et al., 2021). Some studies also mentioned the financial disadvantages of setting up VCMs and the importance of the state’s role in the form of subsidies required to undertake such projects (Dufour et al., 2024; Galgani et al., 2014).

However, what was found to be missing in the existing literature was an article that synthesizes and summarises the vast corpus of studies into a resource base that would benefit future research in the context of the VCMs. Since the VCMs have projects spanning diverse geographies with major existence in a few areas (Global South regions), as revealed by the

Table 1
Types of VCM projects (as identified).

S. No.	Project type category	Project type
1	Renewable Energy	Wind power, hydropower, solar power/heat, geothermal power/heat, biomass power/heat, tidal power, biofuels, green hydrogen
2	Transportation	Fleet efficiency, the modal shift towards EVs
3	Industrial emissions	Cement blending, N2O from adipic acid, SF6 from transformers, HFC23 from HCFC22 production
4	Carbon capture	Carbon Capture and Storage (CCS), Carbon Capture and Utilization (CCU)
5	Waste Management	Landfill gas capture, methane reduction from wastewater, composting, recycling, municipal solid waste (MSW) incineration
6	Agriculture	Manure management, methane reduction from rice cultivation, methane reduction from ruminants, N2O reduction from fertilizer application
7	Land-use change and land management	Afforestation/reforestation, avoided deforestation and forest degradation, improved forest management, peatland restoration, coastal wetland restoration, jurisdictional REDD+
8	Chemical process	Ozone-depleting substances/refrigerants
9	Household and Community	Clean cookstove projects

Source: Authors’ compilation

extant literature, the study categorizes the corpus of articles based on their geographical location for synthesizing. That is, the select studies were first segregated based on the country in which it was conducted. If there were many countries in which the study was conducted, such studies were categorized under multi-country. To further enhance the analysis, the studies from different countries were then clubbed under different climatic zones. The regions falling under similar climatic zones are integrated to derive meaningful insights and add to the existing literature knowledge. Such classification is considered as climate change impacts differ across regions due to unique geographical and climatic conditions. Climatic zones categorize regions based on their weather patterns, which can significantly impact agriculture, energy consumption, health, and infrastructure needs. Analysing data within these zones helps to understand regional variations and their implications (IPCC, 1998).

Hence, an attempt has been made to streamline the past literature and conduct a review focussed on answering the following questions:

RQ1. What meaningful and critical insights can be drawn from the extant empirical evidence-based literature in VCMs across the globe considering different climatic zones and geographical locations of the VCM projects?

RQ2. What key themes were discussed in the extant studies and their subsequent evaluation to inform policy-making, and thereby identifying new areas of exploration?

RQ3. What are the key methodologies employed in the extant studies and how has the empirical evidence-based research in this domain evolved with time?

The objective of this study is to answer the above-mentioned research questions. Hence the study has been divided into five sections. The first section was the introduction section that critically assessed the VCM and highlighted the importance of conducting a study in this domain. It provided the groundwork for the identification of the knowledge gap thus the above-mentioned research questions. The second section discusses the research methodology adopted in this study to answer the enumerated research questions. The third section catalogues the findings of the review of extant literature pertinent to VCMs. It is further divided into sub-parts (Common findings from the selected studies, Findings based on climatic classification, Findings based on the regional classification, drawing of the key themes from the extant literature, research timeline, and methods) used in the studies for a systematic and organized approach to answering the research questions. Additionally, the study digs into the gendered dimension in certain VCM projects. The policy recommendations, considerably anchored in evidence, have been attempted in the fourth section of this study. The fifth section states the key research gaps and observed inconsistencies in findings from the extant literature, and the sixth section enumerates the conclusion of this study and gives possible future research directions in this domain.

2. Methods

The study conducted a systematic review to synthesize the current literature on VCMs. It entails a methodical search for studies and strives for a transparent description of study identification, the rationale for selecting the studies for this review, and how the inferences drawn relate to the pertinent literature. The study followed the guidelines and approach to systematic literature search and review from the work of Cooper et al. (2018), which is detailed below.

2.1. Search strategy

The study used various research databases such as Scopus, Web of Science, Wiley, Taylor and Francis, Springer Nature, and Google Scholar to conduct a comprehensive search. The search terms included were

“carbon market” and “carbon credit market” for the first round of filtration. After going through the articles that appeared, this study refined the search even further and hence modified the search string to “voluntary carbon market”, “carbon credit”, and “carbon credit market” combined with the Boolean operator “OR”. The time frame (in the filters section) of the search results was not restricted so that the third research question could be answered. To enhance the scope of this study, the role of the COP in streamlining the VCMs policy framework was incorporated, hence the search query “voluntary carbon market” AND “COP” was added. The search resulted in the identification of 223 Journal articles, book chapters, and papers from conference proceedings from all the databases mentioned above (The breakdown of the articles as collocated from all the databases in an Excel sheet is available with the authors).

2.2. Inclusion- exclusion criteria

The language of the studies was limited to “English” and the subject area to Social Sciences, Environmental Studies, Agriculture, and Business and Management considering the domain area of our research (socio-economic and policy perspective). Book chapters and papers from conference proceedings were included for preliminary investigation only. This helped develop an idea about the ongoing work in this area and gain contextual clarity. However, they were marked with a note to recognize such studies later in case of elimination. Thus, initially, the study tried to include all the results to understand the subject matter exhaustively. One of the most important criteria for the inclusion of a study was its ability to provide insights based on empirical evidence.

2.3. Study selection process

Similar articles (with the same Titles and Authors) across databases were excluded before the identification. The preliminary screening of these 223 identified articles was conducted after reading their Titles and Abstracts. A separate colour code was then assigned to the studies based on their nature. For instance, empirical evidence on VCM was coded Blue, contextual and descriptive papers were coded Cyan, and so on three other colour codes were defined. The main motive behind this exercise was to (1) segregate Journal articles from Book chapters, and Conference proceedings, (2) identify studies based on empirical evidence or otherwise, and (3) highlight the prime focus of the selected body of literature. After performing this exercise iteratively following the dual-review approach to minimise study selection bias (Stoll et al., 2019), all the articles from conference proceedings along with other articles found to be irrelevant were eliminated. Thus, 91 empirical evidence-based journal articles and book chapters for full-text review were left.

The selected studies were then scanned for specific mention of (1) the “Voluntary Carbon Market”, (2) the empirical evidence component, and (3) a careful assessment of the overall study. The full-text reviewing stage involved individual level scrutiny by the co-authors followed by collective scrutiny to ensure all relevant studies were included in the selection. Finally, sixty-three studies were considered for the analysis, as depicted in Fig. 1. This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) approach to report the inclusion-exclusion criteria to ensure transparency and minimise bias. Also, the quality of the articles selected for the review was carefully evaluated, and as a result, only those articles published in journals indexed in Scopus/Web of Science were considered; articles from non-indexed journals were excluded.

While developing a broader understanding of the subject matter, the work of Lehmann (2019), which focused on gendered dimensions in VCM projects criticized the narratives around VCM projects (cook-stove-based) for promoting gender inequality. Although it did not answer the proposed research questions of this study directly, the narrative encouraged the authors to delve deeper into such critical

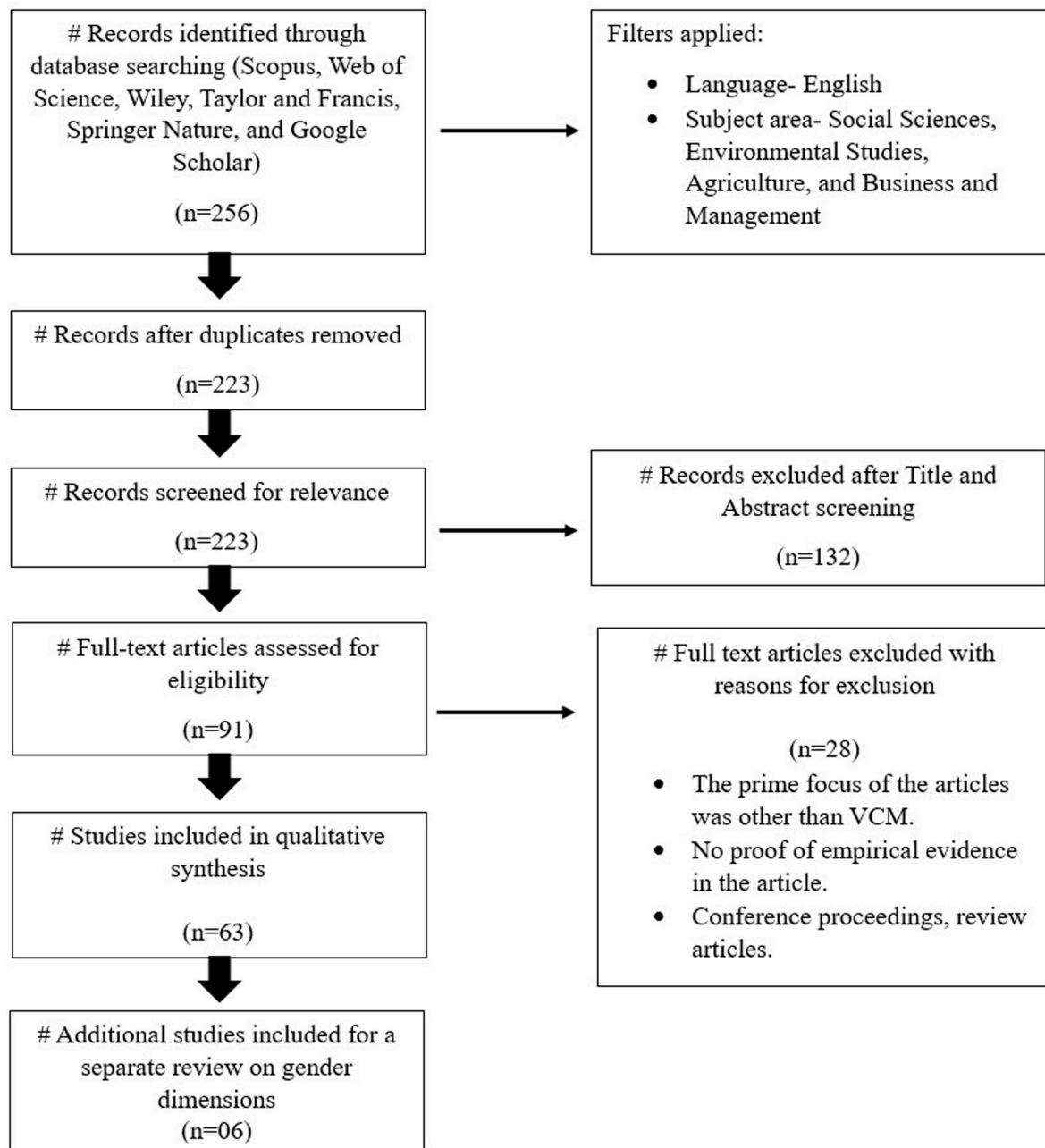


Fig. 1. Flowchart depicting the selection of studies for the systematic review [Source: Authors' compilation based on the methodology].

aspects. Therefore, another search string “Women” AND “Voluntary Carbon Markets” on Google Scholar, Web of Science, and Scopus was run to look for empirical evidence-based articles in VCM specific to gender-related issues. Thus, 6 more studies were found suitable for additional review (recorded in a separate section) along with our main review, which increased the total of the selected studies to sixty-nine.

In order to ensure transparency and methodological rigor, a structured quality assessment was conducted for all sixty-nine studies included in this systematic review. A scoring framework was developed by the authors, guided by methodological recommendations from [Lib-erati et al. \(2009\)](#) and on the methodological and reporting quality of systematic reviews by [Tian et al., 2018](#). The scoring details are presented in [Appendix I\(A\) and I\(B\)](#).

2.4. Data analysis

Data was extracted from the sixty-three studies in a standardized Excel sheet (available with the authors) after thoroughly reviewing them. The key components considered included the (1) Title of the study and the author(s), (2) Specific scope of the study, (3) Geographical location of the study, (4) Methods used in the study, and (5) Key findings of the study. Subsequently, the articles were examined in detail, attempting to understand the overall outcomes and what key inferences can be drawn.

To address the first research question, the studies were categorized into climatic zones based on the Koppen classification system. It is one of the most commonly used climate zone classification methods created in 1884. Although it has undergone numerous modifications over time, scientists worldwide continue to use it ([Met Office, 2025](#)). The

classification process and comparison of studies across climate zones and regions presented several challenges. These included cases where a single country (study location) encompassed multiple climatic zones; studies that lacked specifically defined project locations due to their cross-border nature; and studies with clearly defined locations that nonetheless spanned multiple climatic regions. To address these issues, exact project locations were identified wherever possible, and their corresponding climates were recorded. Additionally, a separate table was created for studies that could not be classified under any specific climatic zone, and another group was formed for studies that belonged to multiple climatic zones (Section 3.2). Another challenge worth highlighting here is the collection of data related to VCM projects from different regions, particularly in the regions with underdeveloped infrastructure and logistics. This is more so in the context of developing countries, where data quality and integrity are often a challenge. Such challenges would be further exacerbated by challenges specific to VCM projects such as additionality, permanence, leakage, and location of project (Pan et al., 2022), accurate measurement of carbon credits, transparency (Wongpiyabovorn et al., 2023), and gaining trust of the stakeholders (Tanveer et al., 2024).

Nevertheless, the key findings from the selected studies (for this systematic review) falling into different climatic zones or groups were subsequently examined (Table 3, and corresponding detailed Tables in Appendix II-VII). For this, the geographical locations of the studies were marked to classify them into the categories of Global North and Global South (Fig. 2). The analysis has been reported in the Findings and Discussion section (Section 3.2) to answer the first research question (RQ1). The whole corpus of studies was then synthesized to draw broad themes (Fig. 3). Based on the results, the important themes were identified for detailed investigation (Section 3.3), and the ideas discovered were used to suggest policy recommendations and highlight future research scope in subsequent sections. This helped the authors to address the second research question (RQ2). Finally, the methods employed by the studies and the evolution of research over time in the VCM domain were analysed, and the results were reported (Section 3.4), which helped to answer the third research question (RQ3).

A breakdown of the selected articles for the review is enlisted in Table 2. From the studies, the majority (thirty-one studies) fall into the Tropical climatic zone. An assessment reveals that the locations of the projects in this zone are primarily the Global South regions (Asia and Africa). The projects implemented here are mostly forestry projects which can be attributable to the climatic conditions (tropical rainforest, sub-tropical areas) here. Following the tropical zone, the temperate zone has fifteen studies that belong to multiple regions (Europe, New Zealand, USA). The continental zone has seven projects which are mostly in the Global North of the world and hence the nature of these studies is different from that in the other zones (mentioned in the subsequent sections). Due to extreme weather conditions, no studies specific to VCM are found in the Tundra region (detailed breakdown available with the authors).

Table 2
Breakdown of the studies selected for the review.

Basis of classification	Number of studies
Climate Zones	
Tropical	31
Temperate	15
Arid	3
Continental	7
Tundra	0
Multiple climate zones	4
Others	
Multi-country studies	3
Women and VCM	6
	69

Source: Authors' compilation

Table 3
Aggregation of findings from the selected studies.

Climate Zone (No. of total studies)	Themes discussed	No. of studies	Authors	Reference Table for details
Tropical Zone (31)	Land use	5	Pinnschmidt et al. (2023); Naime et al. (2020); Cameron et al. (2019); Benson et al. (2017); Grace et al. (2010)	Appendix II in Appendix File
	Governance	4	Garcia et al. (2021); To et al. (2017); Paiva et al. (2014); Pettenella and Brotto (2012)	
	VCM project ROI	6	Singh et al. (2024); Santhiyami and Suparti (2023); Kongmeesup and Boonyanuphap (2019); Galgani et al. (2014); Kalaba et al. (2013); Butler et al. (2009)	
	Policy for VCM	4	Zhang et al. (2024); Indrajaya et al. (2024); Boer (2018); Martello et al. (2010)	
	Carbon finance	1	Cong et al. (2021)	
	Forest Management	1	Lupala et al. (2017)	
	Co-benefits	3	Lock (2024); Paiva et al. (2014); Lasco et al. (2010)	
	Blended finance	1	Aerni (2016)	
	Carbon storage & assessment	3	Somprasong (2024); Bomfim et al. (2022); Chayaporn et al. (2021)	
	Sustainable Agriculture	2	Cariappa et al. (2024); Cariappa et al. (2024)	
	Climate Justice	2	Kent and Hannay (2020); Duker et al. (2019)	
Temperate Zone (15)	Forest Management	3	Ezquerro et al. (2024); Nonini and Fiala (2021); Moreno et al. (2019)	Appendix III in Appendix File
	Carbon storage & assessment	1	Cuesta et al. (2023)	
	Blended finance	1	Moxey et al. (2021)	
	Policy for VCM	2	Meitern (2024); Birchall et al. (2018);	
	Governance	1	Sharma et al. (2023)	
	Carbon finance	1	Ranjan (2024)	
	Co-benefits	2	Zheng and Zhang (2023); Birchall et al. (2018)	
	Carbon footprint	1	Almeida et al. (2014)	
	VCM project ROI	1	Dwivedi et al. (2009)	

(continued on next page)

Table 3 (continued)

Climate Zone (No. of total studies)	Themes discussed	No. of studies	Authors	Reference Table for details
Arid Zone (3)	Citizens' views on VCM	1	Thompson and Hansen (2012)	Appendix IV in Appendix File
	Land use	1	Li and Martino (2024)	
	Policy for VCM	1	Cammarata et al. (2024)	
	Land use VCM project ROI	1	Plieninger (2011) McHale et al. (2007)	
Continental Zone (7)	Carbon storage & assessment	1	Stivins et al. (2024)	Appendix V in Appendix File
	Citizens' views on VCM	3	Karhunmaa et al. (2023); Kuhn and Uler (2019); Miller et al. (2014)	
	Policy for VCM	1	Dufour et al. (2024)	
	Co-benefits	1	Park and Oh (2016)	
	VCM project ROI	1	Contasti et al. (2023)	
Multiple climatic zones (4)	Land use	1	Aneseyee et al. (2022)	Appendix VI in Appendix File
	VCM project ROI	2	Watson et al. (2013); Karavai and Hinostroza (2013)	
	Climate justice	1	Mathur et al. (2014)	
Multi-country devoid of specific climate zone (3)	Co-benefits	1	Teo (2024)	Appendix VII in Appendix File
	VCM project ROI	2	Aryal et al. (2024); Valiergue and Ehrenstein (2023)	

Source: Authors' compilation based on extant literature [detailed version available in Tables 1–6 in Supplementary File 1]

3. Findings and discussion: contribution to the existing literature

3.1. Findings common to most of the studies

A detailed study of the key findings reveals that the chosen articles cover various themes spanning from discussing potential financing sources to assessing the economic viability of the projects with essential touch points such as climate justice and sustainable land use being studied among others (Fig. 3). Some key findings related to economic viability assessment and probable sources of financing for the carbon projects remain commonly observed. For instance, most stakeholders of VCM projects (landholders, project certifiers, project owners, and project beneficiaries) emphasize its economic viability overshadowing the social and environmental aspects when choosing a potential project to invest in (Karavai and Hinostroza, 2013; Naime et al., 2020; Butler et al., 2009). The economic viability, assessed through various indicators such as Internal Rate of Return (IRR), Returns on Investment (ROI), Net Present Value (NPV), Index benefit/cost (IBC), and Index ROIA/Minimum Rate of Attractiveness (MRA), is indicative of the profitability of a project (Butler et al., 2009; De Lima et al., 2015). Remarkably, fifteen articles from the chosen body of literature investigate the economic viability of VCM projects by employing the ROI parameter. A few articles among these, further propose various means to enhance the ROI from VCM. For instance, the inclusion of sustainable practices and techniques in projects related to forest management (Ezquerro et al., 2024; Nonini and Fiala, 2021) and agriculture (Cariappa et al., 2024). Further, investment in green electricity production plants generates quicker and more positive ROIs than other VCM alternatives such as forest conservation, cleaner cooking through cookstoves, sustainable agricultural practices, and waste management (McHale et al., 2007). Apart from the ROI aspect, the studies concerned with the potential sources of finance for carbon projects peg state funding in VCM as crucial as it facilitates expansion and motivates private players to participate actively in climate change mitigation through VCM mechanism (Sharma et al., 2023).

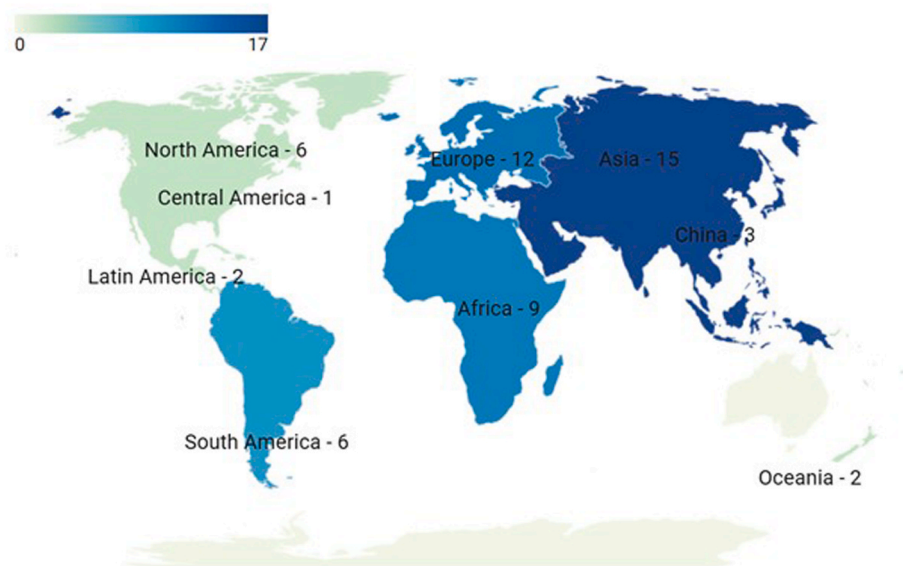


Fig. 2. Studies conducted across different geographies including Global North and Global South regions

[Figure Source: Authors' compilation with the help of online software- Data wrapper [In addition to the above studies, there are four studies which are multi-country but the locations are unidentifiable and six studies which discuss the Gender dimension in VCM.].

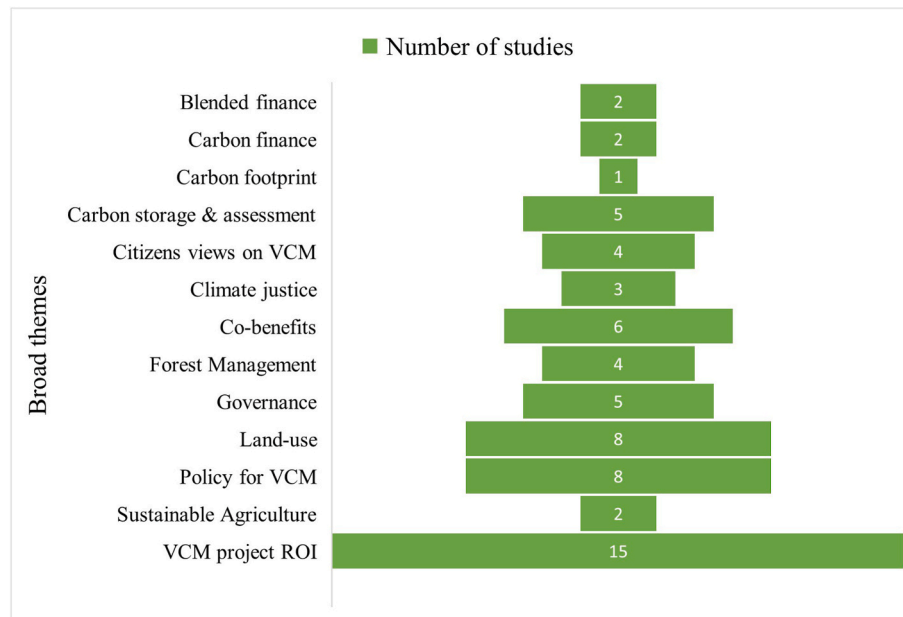


Fig. 3. Broad themes of the research articles
[Figure Source: Authors' compilation].

3.2. Findings based on the climatic classification and geographical locations

3.2.1. Studies specific to tropical climatic zone

In the case of climatic zone classification, considerable variations among the key findings can be noted across various themes and the outcomes of the studies. For instance, while analysing Table 3 (and Appendix II), it can be seen that studies specific to the tropical zone primarily investigate the forestry sector. Further assessment of the geographical locations as stated in Appendix II reveals that most of the study locations are in the countries in the Amazon basin (e.g., Brazil, Peru, Colombia, and the Amazon rainforest) and Indonesia in Asia. Pertinent to the tropical climatic zone, a few studies examining the economic viability of Reducing Emissions from Deforestation and Forest Degradation (REDD) projects linked to VCM found that such projects are not financially feasible (Duker et al., 2019; Garcia et al., 2021). For similar reasons, other studies emphasized economically viable alternative land use such as palm oil production (Butler et al., 2009) or agroforestry (Santhyami and Suparti, 2023; Singh et al., 2024). Yet, a few studies suggest REDD VCM projects as a potential additional income source for small farmers, particularly when Community-based Forest Management (CBFM) activity is approved as a REDD + option (Lasco et al., 2010). Remarkably, CBFM projects linked to VCM are also likely to improve social capital and economic viability (Lupala et al., 2017). Nevertheless, irrespective of the contrasting outcomes related to the REDD projects' financial feasibility, almost all studies concerned with REDD point towards governance issues. Therefore, an effective framework is required to strengthen the governance of such projects (Garcia et al., 2021). This will not only clearly define the role of the private and public sectors in such projects (Pettenella and Broto, 2012) but will also help to combat misalignment between the project's intended outcomes and local interests (To et al., 2017). Policymakers can potentially focus on enhancing digital intelligence infrastructure to promote low-carbon transformation; optimizing market structures through appropriate market segmentation to encourage competition and technological innovation; strengthening governance and environmental regulatory capacities to ensure the effectiveness of policy implementation; and balancing market integration to prevent environmental regulations from being weakened by market unification (Yue and Han, 2025).

Another observation pertinent to the tropical climate is related to

confrontational findings regarding certain plantations offering a better return on investment than their monocultures. While Pinnschmidt et al. (2023) suggested that mixed-species plantations outperform their monocultures when carbon payments are concerned, Santhyami and Suparti (2023) found that monoculture practices have dual benefits of carbon offset and economic benefits in the case of cocoa plantations. This could be attributed to different geographical locations and the type of plantation being observed and reported. For instance, the tropical mixed-species plantations investigated in Costa Rica fall in the Central America region while cocoa plantations have been examined in West Sumatra, Indonesia in Asia.

Further assessment reveals that some studies in the tropical zone focus on the carbon storage potential of various projects and their ability to issue carbon credits. These usually employ remote allometric modelling (Chayaporn et al., 2021), or scenario analysis (Pinnschmidt et al., 2023) (Section 3.4) to estimate carbon storage in Teak plantations sites. Similarly, forest Miombo woodlands (in Africa) (Kalaba et al., 2013; Benson et al., 2017), mangrove plantations (in Asia) (Cameron et al., 2019), and projects related to composting and biochar sequestration (Galgani et al., 2014) also estimate carbon storage. It gives some project stakeholders, namely owners, certifiers, and beneficiaries, verified information related to the carbon sequestration potential of the projects. This information along with the economic viability aspects of the potential VCM projects allows the stakeholders to make informed decisions regarding their investments.

While analysing the type of studies conducted, it is worth noting that only a limited number of studies address the issue of climate justice. The host communities responsible for the implementation and operations of VCMs have little to no knowledge regarding their importance in aiding climate change and receive no considerable benefits from contributing to such projects. This sheds light on the grey area surrounding the investments linked to VCMs and raises concerns regarding justice for the local communities. Although these issues can be addressed to some extent through proper communication of such schemes to the local communities (Kent and Hannay, 2020) and instilling ownership in them (Martello et al., 2010) future work can focus on developing frameworks to address the issue that act as barriers to the empowerment of the communities involved.

3.2.2. Studies specific to temperate climatic zone

The studies specific to the temperate zone discuss varied themes across various regions as seen in Table 3 (and Appendix III). These themes broadly cover forest management, investment, economic viability, and co-benefits of such projects. Interestingly, discussing the context of forest management and its attractiveness for VCMs, a few studies considered the agency of private land owners. While looking at the outcomes from different regions (North America, Latin America, Europe, and New Zealand), certain regions (for example in the US) showed that forestry projects could be profitable (Dwivedi et al., 2009). However, outcomes from other studies in the USA suggested that the landowners may have a passive approach toward undertaking voluntary carbon sequestration activities until regulatory supervision (Thompson and Hansen, 2012). Essentially, a requirement for clear rules surrounding VCM, both regionally (Meitern, 2024) and internationally, needs urgent attention (Thompson and Hansen, 2012). Further, a few studies also cite the importance of the inclusion of Integral Forest Management (IFM) in Strategic Forest Management to achieve economic viability. For instance, IFM (timber resources and carbon storage) linked to VCM poses an attractive option for private players (Moreno et al., 2019) and reduces uncertainties regarding the development of carbon markets (Cuesta et al., 2023).

In the European region, the VCM projects were seen to have a comparatively higher opportunity cost of implementation. Opportunity cost refers to the value of the best alternative forgone encompassing not just monetary costs but also time and utility lost by not pursuing other options (Riera-Prunera, 2022). People normally do not pay sufficient attention to opportunity costs and as a result make poorer and less efficient decisions (Maguire et al., 2023). It is further observed that the opportunity costs farmers face has a role to play in designing sound policy incentives (Wale, 2007). Thus, blended finance (private and public) as a source of financing is considerably important in VCM (Moxey et al., 2021). The private sector funds usually play a major role in compensating farmers for undertaking VCM activities (Li and Martino, 2024).

In New Zealand, the development of a framework for VCMs is discussed which shows early success in the market including shifted behaviour of private players around climate change mitigation. However, the growth of VCM has lately slowed due to external factors including environmental variations, influence of large organizations, communication challenges, low certification recognition, risk of green-wash exposure, policy uncertainty, the global financial crisis, and general disenchantment with the carbon market. This has led to market stagnation with buyers and sellers drifting away from the market (Birchall et al., 2018).

In China, the growth of VCM has depicted several regional spillover effects including city greening (Zheng and Zhang, 2023). This is primarily owed to the encouragement surrounding the development of forest carbon sequestration projects. The forestry-based carbon projects linked to VCM not only improve farmers' income considerably but also have the potential to generate employment and develop social capital (Grace et al., 2010).

3.2.3. Studies specific to arid and continental climatic zones

A limited number of studies were conducted in the Arid climatic zone broadly covering agroforestry and the agriculture sector (Table 3) (detailed breakdown in Appendix IV). The agroforestry projects in North America (Plieninger, 2011) and urban tree planting projects in Europe (McHale et al., 2007) are found to be a cost-effective means of sequestering carbon. Further, studies examining the agriculture sector emphasize offering extension services and technical assistance to help farmers understand the potential of the VCM (Cammarata et al., 2024).

The studies specific to the continental zone are mostly conducted in developed countries (Appendix V). Individual responsibility towards voluntary carbon offsetting seems to be considerably missing among the citizens of Finland due to low trust in the concept of VCM (Karhunmaa

et al., 2023). American citizens are aware of climate change and believe in taking steps towards its mitigation (Kuhn and Uler, 2019). Therefore, the total offset purchases increase with an increase in the total damages. Further, a few studies using cost-benefit analysis of carbon credit offsets show that current carbon prices are not likely enough to encourage landowners to participate in the carbon markets (Miller et al., 2014) and that carbon credit income is economically promising only while supplementing the main income (Contasti et al., 2023).

Peatland ecosystem is often seen as a potential option for developing VCM projects in continental regions because of their climate change mitigation potential (Bonn et al., 2014). Examining peatland projects, Stivins et al. (2024) mention better techniques such as rewetting in the case of Peatland to increase carbon storage leading to better carbon prices. Policymakers should also consider latent expenditures (a few being opportunity cost, transaction cost, transport cost, and implementation cost) while adopting the REDD mechanism in the future, as suggested by Park and Oh (2016) based on a study in North Korea.

3.2.4. Studies falling in multiple climatic zones

In addition to the findings mentioned in the first section above, a few studies spread across multiple climatic zones also highlight changes in carbon storage as a result of changed land use (Appendix VI). For instance, the work of Aneseyee et al. (2022) spread across temperate and tropical regions in Africa, suggests that the highest loss of carbon storage and valuation is found in forest land followed by grazing land. When it comes to the rewards from VCM projects to its beneficiaries, Mathur et al. (2014), through case studies across Asia and Africa covering multiple climate zones show that host communities benefit little from such projects. This situation arises as the carbon market focuses on the cost-effectiveness of VCM projects overshadowing other aspects (Karavai and Hinostroza, 2013).

Furthermore, few studies could not be categorized into any of the climatic zones since the focus is on multi-country data collection (Appendix VII). Some of these studies highlight the need to make offsets and emissions commensurable due to differences in opinion in terms of the importance of co-benefits to the market players (Teo, 2024). The co-benefits, also seem to be lacking considerably in terms of quality and integrity serving as barriers for retailers to go ahead with VCM projects (Valiergue and Ehrenstein, 2023). Further, the REDD + mechanism is questioned with advice to secure resource ownership and management authorship of locals to make it a successful VCM project (Aryal et al., 2024).

3.2.5. Global north v/s global south: the non-geographical divide

A continental classification of the study locations (Fig. 2) depicts that the maximum number of studies are conducted in Asia followed by Europe. Further breakdown reveals that Indonesia (in Asia) is the primary hub of research studies in the VCM. This can be attributed to the concrete steps undertaken by the Indonesian government to save forests, initiate REDD+, and communicate its Nationally Determined Contribution to aid climate change mitigation after COP 13 culmination in Bali (Republic of Indonesia, 2022). The studies here are carried out in the forestry sector predominantly surrounding the REDD + initiative.

Other noteworthy locations in Asia are Thailand, India, and China. China has been classified into the Global South by its president but it is an emerging superpower (Brooke-Holland, 2024). The Global North-Global South divide is non-geographical but is used to classify nations based on their social and economic development and the level of industrialization. It has become evident in this era of climate change owing to the disparity in the destruction caused by the two groups of nations. The Global South has often debated, as was recently mentioned in the COP- 29 as well, that global climate change has been caused by the massive industrialization of the Global North nations, and hence they should be penalized more for the damages. This includes not only regional remedies to be implemented by the Global North but also the disbursement of financial compensation to the Global South to divert

land use towards projects that can aid in carbon sequestration from the atmosphere (United Nations, 2023).

One of the biggest outcomes achieved during the COP-29 proceedings was the final agreement on the rules under Article 6 of the Paris Agreement, which will help countries and companies trade carbon credits more transparently and fairly. A global registry was also announced to track carbon credits and avoid any double-counting. Further, developed countries promised to provide \$300 billion each year by 2035 to support climate projects, including those in developing countries. There was a strong push to make sure that carbon credits are high in quality and help reduce emissions. These changes aim to make carbon markets more reliable, fair, and accessible, especially for countries in the Global South (UNFCCC, 2025).

Proceeding further, owing to the north-south divide, a higher concentration of the studies in the Global South regions was seen because of several VCM projects in these locations (owed to lower marginal cost and low GDP) (Cong et al., 2021). On the other hand, the studies conducted in the Oceania and North American regions of the Global North typically focused on citizens' views related to carbon emissions and offsetting (Miller et al., 2014; Kuhn and Uler, 2019), the sources of carbon emissions in modern-day organizations (Almeida et al., 2014), and discussed the evolution of carbon markets in specific regions (Birchall et al., 2018). However, Europe, which has the most number of studies in the Global North, had some studies that investigated and reported governance issues in forest management (Sharma et al., 2023), financing options for carbon projects (Meitern, 2024), and land use management scenarios for various agricultural techniques (Stivrins et al., 2024; Plieninger, 2011).

Hence, the pattern of the studies selected for the review (types of projects, intended research objectives, outcomes, and location of the studies) considerably mirrors the non-geographical divide.

3.3. Key themes drawn from the selected studies

The major focus of most of the studies is on calculating the returns on investment of the VCM project under consideration. While few studies compute the returns of the projects to classify them as profitable or loss-making (Santhyami and Suparti, 2023; Dwivedi et al., 2009), several studies include a comparative analysis of the projects to portray alternative land use that is economically viable. For example, mangrove rehabilitation over disused ponds (Cameron et al., 2019) pastures and young forest over old forest (Naime et al., 2020), abandoned saltmarshes over agriculture (Li and Martino, 2024), and palm oil cultivation over REDD forest (Butler et al., 2009). Another set of studies investigates the Carbon storage and assessment of land being used in various forms to report maximum carbon storage and thus an increase in carbon credit generation to report optimal usage (Chayaporn et al., 2021; Plieninger, 2011).

The co-benefits of VCM projects are discussed and lauded by a few (Paiva et al., 2014; Lasco et al., 2010) but are also criticized due to their hollow nature (Lehmann, 2019). These are additional benefits resulting from the implementation of VCMs and play an important role in the sale of carbon credits generated through such projects (McHale et al., 2007). It can be inferred that credits with potential for co-benefits (e.g. fostered corporate social responsibility, social cohesion of local communities and voluntary leadership, and positive environmental impacts) are preferred to those focusing exclusively on emission reduction in the VCM. The findings of this study suggest that developing co-benefits is important for strengthening the market competitiveness of forest carbon credits in the VCM (Lee et al., 2015).

The inclusion of sustainable forest management is the prime focus of quite a few research articles (Ezquerro et al., 2024; Moreno et al., 2019) with many raising the issue of the lack of a proper governance framework (Sharma et al., 2023; Pettenella and Broto, 2012). Thus, all the above and other mentioned themes (Fig. 3) highlight that VCMs are still in their developing stage across the globe. Immediately upon the

accomplishment of fundamental requirements such as the formation of a policy framework for setting up VCMs, the release of guidelines for carbon trading, and a description of ethical dimensions of the activities in the VCM value chain, they show a predisposition towards fulfilling the underlying cause for their establishment.

3.3.1. Returns from VCM projects – juxtaposing global south and global north

Studies examining returns or economic viability of VCM projects in the global south suggests mixed outcomes. For example, agroforestry linked to VCM showed a potential of 40–50 percent increase in income to farmers in India (Singh et al., 2024), and VCM linked cocoa-based agroforestry system showed greater returns in Indonesia (Santhyami and Suparti, 2023). Similarly, study from Thailand (Kongmeesup and Boonyanuphap, 2019) and Zambia (Kalaba et al., 2013) highlight the potential of greater returns through carbon storage in woodland ecosystem. In a contrary, a few other studies highlight the limitations of projects linked to VCMs to provide expected returns. For example, carbon markets can make biochar projects viable only at certain threshold price of the carbon credits, as found in Ghana (Galgani et al., 2014). Similarly, a study in Indonesia highlights how alternative land uses could be more attractive investment options of the carbon market projects if VCMs fails to incentivise the investors (Butler et al., 2009). Further, studies concern uncertainties in carbon pricing in VCM would impact the financial decisions and expected returns (Watson et al., 2013).

There are a relatively few studies from the global north providing insights related to returns from the VCM projects or their economic viability. These few studies from the United States also offers a mixed idea, like the studies from the global south. A study in southern United States clearly suggests better returns for non-industrial private forest landowners from VCM (Dwivedi et al., 2009). However, a study conducted in Colorado reveals high upfront cost and long-term commitment posed barriers to realizing immediate returns from afforestation efforts despite long-term potential gains (McHale et al., 2007). Similarly, a study in mid-southern United States reveals potential trade-off between income from carbon markets and yield income for soybean farmers, and the role of various factors on net income.

3.3.2. Studies about land use in VCM

Land as an asset is limited on earth which explains its appreciation in value over time. The VCM projects that aid in mitigating climate change involve the use of land in several forms. For example, it can be used in Agricultural projects practicing carbon farming or in forestry projects bound by the REDD + mechanism. Therefore, studying the effectiveness of land use in different VCM projects becomes eminent to ensure sustainable use of the asset. From the extant literature, eight empirical studies specifically focus on the land-use aspect of the VCM. A few among these computes the economic viability of the ongoing projects and hint towards better land usage alternatives. For instance, Pinnschmidt et al. (2023), while studying an agriculture project linked to VCM, found that tropical mixed-species plantations can outperform monocultures in terms of carbon sequestration and economic return. Naime et al. (2020) assess the economic valuation of ecosystem services and find that landholders prefer to buy young forests over old forests (which can sequester more carbon) as the former can be converted into pastures that are economically beneficial. Li and Martino (2024) investigate the economic feasibility of VCMs in land use management scenarios for Scottish saltmarshes. Further, some studies focus on examining the carbon storage potential of land in its current form of use, for example, Benson et al. (2017) and Cameron et al. (2019) assess the mangrove plantations; Grace et al. (2010) examine the African woodlands; and Plieninger (2011) studies agroforestry projects. State intervention in the form of management and regulation of land use is crucial for reducing the expansion of agricultural areas to mitigate negative impacts on the ecosystem, as found in the case of oil palm plantations and aquaculture

by Rattanarat et al. (2024). Along with the prime focus listed above, many studies cite alternative land uses as well to inform policy-making for better decisions. For instance, an inclination towards agroforestry projects as opposed to the REDD mechanism (Santhiyami and Suparti, 2023; Singh et al., 2024; Plieninger, 2011); promotion of mangrove rehabilitation through conversion of disused ponds into mangrove plantations for dual benefits of carbon offset and economic benefit (Cameron et al., 2019). Hence, the authors argue that knowledge and practice of sustainable land uses are necessary, as they would contribute to solving these challenges, guaranteeing the food supply for all populations, promoting regional economic development, and contributing to the conservation of natural resources for future generations (Sanchez et al., 2019).

3.4. Research timeline and methods used in extant VCM literature

The first empirical evidence-based research article in the field of VCMs was published in 2007. This article studied the cost-effectiveness of urban tree plantations in terms of the carbon credit market. The findings revealed community forests to be more economically profitable and a better marketable solution in place of urban tree planting projects. However, urban tree projects received investment only for the co-benefits associated with them (McHale et al., 2007). The trend of computing the return on investment of such projects continued until 2010 when the requirement of a policy framework for VCMs was highlighted (Martello et al., 2010). Subsequently, the areas of governance (Pettenella and Brotto, 2012) and effective land use (Plieninger, 2011) were addressed beginning in 2011. The area of climate justice was highlighted only in the year 2019 with authors citing the lack of benefit of VCM projects for the host communities, for example, in the context of communicating the concept of “carbon” to local communities (Kent and Hannay, 2020), or exclusion of smallholders from carbon forestry projects (Duker et al., 2019).

The number of studies in this domain was limited to a few per year before a leap was noticed beginning in 2023. The maximum number of studies to date have been conducted in 2024 covering various themes from Sustainable agriculture to Carbon finance (Fig. 4). This can be attributed to the COP meeting agendas focussed on climate change specifically VCM and Climate finance in 2024 (Table 4) (ClimatePartner, 2024). The increasing number of research articles on VCM in the

Table 4

COP themes over the years.

Year	COP Theme
2015	Paris Agreement Adoption
2016	Implementation and Action
2017	Fiji Presidency Dialogue
2018	Katowice Rulebook finalisation
2019	Ambition and delay
2020	Covid year
2021	Net zero pledges
2022	Loss and Damage
2023	Global Stocktake assessment
2024	VCM, Climate Finance

Source: Authors' compilation

academic field depicts the growing scope of this domain and the promising outcomes it provides in climate change mitigation. At the same time, the efforts and commitments of governments cannot be overlooked, as they actively pursue their Nationally Determined Contributions (NDCs) and strive to achieve net-zero targets in alignment with the goals outlined in the Paris Agreement of 2015.

This study also assesses the research methods employed in the extant literature. Qualitative methods inclusive of interviews, surveys, and content analysis have been the most common methods used. Mixed methods including both Qualitative and Quantitative methods succeed the qualitative methods. Further categorizing, the studies attempting to study carbon storage employ remote sensing techniques (Benson et al., 2017; Cuesta et al., 2023) whereas efficient land use is determined through Scenario Analysis (Butler et al., 2009; Pinnschmidt et al., 2023). Thus, the techniques are applied based on the nature and scope of the respective studies (Table 5). Further, interesting patterns were identified in terms of the methodological approaches. Considerable number of studies in both global south and global north used primary data driven methods, for example interviews (Aerni, 2016; Boer, 2018), field surveys (Watson et al., 2013; Aneseyee et al., 2022; Chayaporn et al., 2021), ecological measurements (Kongmeesup and Boonyanuphap, 2019; Cariappa et al., 2024), and other participatory research approaches (Kent and Hannay, 2020; Lock, 2024) providing rich micro-level understandings. Remarkably, studies in the countries that have made considerable progress in regulating emissions, e.g., China (Zheng and Zhang, 2023; Cong et al., 2021), USA (Dwivedi et al., 2009),

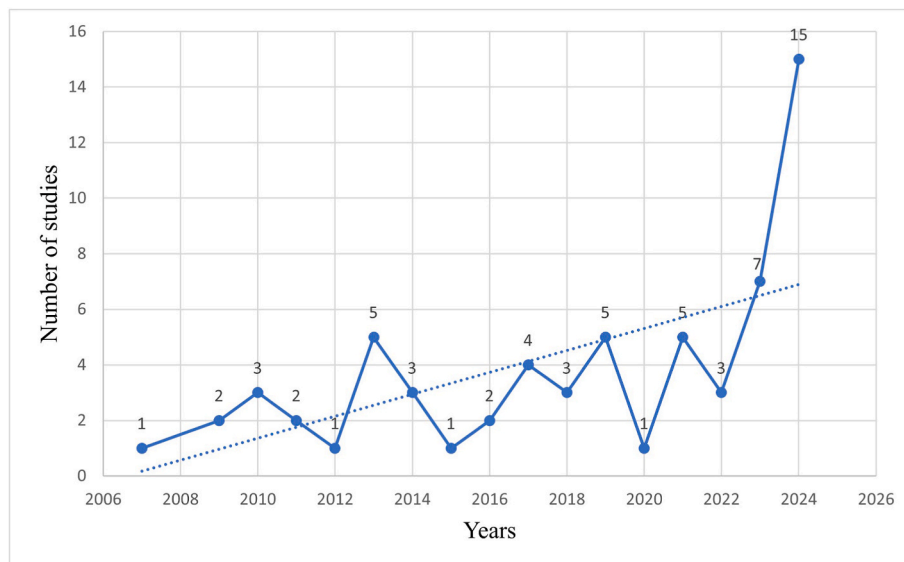


Fig. 4. Empirical evidence-based research timeline in VCM

[Figure Source: Authors' compilation (The Trendline (represented by the dotted line) showcases the constantly increasing empirical-evidence based research articles on VCM)].

Table 5
Research methods used in VCM.

Research Method	Number of studies
Qualitative Analysis	17
Remote Sensing	5
Case study	8
Quantitative	10
Experimental	3
Exploratory	1
Mixed	12
Quasi-experimental	1
Simulation	2
Sensitivity Analysis	1
Scenario Analysis	3

Source: Authors' compilation

South Korea (Kim, 2022) and New Zealand (Birchall et al., 2018), exploited the available national-level secondary databases to provide macro-level understandings. Certain studies also highlighted methodological challenges owed to defining additionality, permanence, and leakage, for example, in India (Cariappa et al., 2024), in Germany (Pleninger, 2011), or multi-country (Wissner et al., 2022), which future studies may consider to address by developing robust methodological framework. More details concerning the methods used, type of data and sources, and case study context are represented in Table 7.

3.5. Studies related to “Women and VCM”

Beyond the stated objectives for this study, the intent of reviewing the studies related to “Women and VCM” is to gauge the role of gender in VCM-based projects (Table 6). This section has been incorporated in addition to the primary review to use gender as an analytical category. Doing so, rather than understanding it as a biological given, allows us to focus on power relations within development schemes to understand under what processes women are left at a disadvantage in relation to men (Gay-Antaki, 2016). Thus, VCM projects, which also have broader implications for the development of communities, are also required to be examined through the gender perspective. As already mentioned in a previous section, this study considered a few research articles that helped to attain certain critical perspectives. Remarkably, projects examining women's role in reducing carbon emissions find that the project marketers/vendors build a “carbon as an asset” narrative to sell VCM project credits at an enhanced value (Lock, 2024), introducing the concept of “charismatic” carbon. They leverage the societal benefits offered by a project and encash the prestige associated with it. The companies buy such projects to fulfil their Corporate Social Responsibility (CSR) and Environmental, Social, and Governance (ESG) compliance. Thus, women's empowerment associated with charismatic carbon gets sold and recorded as an achievement (Wang and Corson, 2015). However, in reality, it undermines the agency of women. It criticizes the aspect of social equality and freedom of choice, which are fundamental to the feminism theory (Lee et al., 2015). The picturization of women with the use of cookstoves highlights traditional stereotypes about their place in society. It contributes to the marginalization of women's contributions outside the home and hinders progress towards gender equality. Although Wissner et al. (2022) suggest optimistic outcomes of the cookstove projects (health benefits and reduced cooking time for women), project developers/certifiers hold the power to change such narratives and they should take this into account.

Yet another study from South Korea, which focused on the governance aspect of corporations, found that a higher proportion of female executives and employees significantly increases voluntary carbon emission disclosures. Such findings highlight the role of women in enhancing corporate transparency and improving the quality of the information environment (Kim, 2022). Hence, the authors argue that women fare better in leadership positions and should be freed from the

Table 6
Women and VCM.

S. No.	Title and Author	Location of the study	Climate zone	Key Findings
1	Implications on equity in agricultural carbon market projects: a gendered analysis of access, decision making, and outcomes (Lee et al., 2015)	Kenya	Tropical/Temperate	Women face limited access and decision-making power in agricultural carbon projects despite contributing more, underscoring the need for gender-inclusive project design.
2	The making of a 'charismatic' carbon credit: clean cookstoves and 'uncooperative' women in western Kenya (Wang and Corson, 2015)	Kenya	Tropical	The cook stove projects funded through carbon offsets bring some benefits to rural Kenyan women, like reduced cooking time and smoke, but rely heavily on their labour while transferring carbon savings profits to Global North investors, marginalizing the women.
3	The Effect of Female Personnel on the Voluntary Disclosure of Carbon Emissions Information (Kim, 2022)	South Korea	Tropical	The study finds that a higher proportion of female executives and employees significantly increases voluntary carbon emission disclosures, highlighting the role of women in enhancing corporate transparency and improving the quality of the information environment.
4	“Now We Have Equality”: A Feminist Political Ecology Analysis of Carbon Markets in Oaxaca, Mexico (Gay-Antaki, 2016)	Mexico	Tropical	Carbon projects exploit women's unpaid labour, exclude them from decision-making, and worsen gender inequalities, while ignoring their vital role in resource management and climate adaptation.
5	Sustainable development impacts of selected project types in the voluntary carbon market (Wissner et al., 2022)	Multi-country	Multi-climate	Such projects reduce fuel collection and cooking time and improve the health of women as compared to traditional cooking methods. This could lead to higher school attendance for children and increase women's ability to pursue income-generating leadership roles.
6	When cultural political economy meets 'charismatic carbon' marketing: A	Asia and Africa	Tropical/Temperate/Arid	Carbon project marketers make strong claims regarding improving

(continued on next page)

Table 6 (continued)

S. No.	Title and Author	Location of the study	Climate zone	Key Findings
	gender-sensitive view on the limitations of Gold Standard cookstove offset projects (Lehmann, 2019)			family livelihoods, income generation, and women empowerment through cookstove-based VCM projects popular in the global south while undermining the influence of women's agency.

Source: Authors' compilation

traditional stereotypical roles to prove their worth while contributing to society's development. The organizations employing women in top positions can capture the symbols of progressive social change and modernity by such appointments (Eagly and Carli, 2003).

4. Policy recommendations

Several studies from the chosen body of literature emphasize the need for regulatory frameworks and guidelines to oversee the VCM mechanism. The role of the government can be seen as of prime importance for the operationalisation of these markets, and for maintaining environmental integrity and co-financing such projects (Boer, 2018; Dufour et al., 2024). Many critical challenges have been identified as major hurdles to the emerging VCM landscape that include concerns related to governance of VCMs, additionality in carbon credits, transparency of project location, beneficiaries, and flow of funds, and market exploitation (through over-crediting, greenwashing, poor-quality projects). Based on the review of the literature and reliable reports, this study provides certain policy recommendations, ranging from global to national level, that are likely to help the stakeholders to address the aforementioned concerns.

- Integrating global governance mechanism with local actors (Paiva et al., 2014), encouraging participatory governance including clearly defined property rights of the communities (To et al., 2017; Sharma et al., 2023).
- As financial flow is mostly controlled at the registry level with a few intermediaries controlling the flow of VCM funds, stringent guidelines encouraging disclosure of information to the credit buyers would be crucial to bring transparency in flow of funds (CCAG, 2024).
- The priorities of countries to participate in VCMs, especially the countries in the global south varies. Hence, the design of regulations needs to be developed to synergize the local concerns/priorities and transnational climate goals (Andonova et al., 2019).
- Use of technology, for example digital monitoring, reporting, and verification (Christiansen, 2024), blockchain technology to enhance traceability and reduce risk of double counting (Boumaiza and Maher, 2024), use of satellite imagery and remote sensing (Pande and I4I Team, 2025), will bring transparency regarding project locations, beneficiaries, flow of funds, and address additionality.
- Requisite reporting of benefit sharing arrangements between project co-ordinators and local communities for transparency, for example some of the projects registered under Plan Vivo (Healy et al., 2023).
- Robust and standardised methodologies to define baselines and carbon offsets required urgently to address issues concerning additionality, as serious discrepancies has been observed in baselines across different methodologies (West et al., 2024).
- Prioritizing the development of a network of actors with carefully defined roles and established trustworthiness would be crucial for

success of projects, particularly in forestry and land use projects (Pettenella and Brotto, 2012).

5. Appraising the key research gaps addressed and inconsistencies in findings

The selected cohort of the sixty-nine studies addressed several critical research gaps surrounding carbon projects (linked to VCMs) using different research methods within diverse case study contexts, and a mix of studies with spatial and temporal coverage. While each study bring its own strengths and weaknesses, the collective cohort of studies across several countries led to rich findings, imparting depth and breadth to our understandings on VCMs and related aspects. A more detailed representation of the identified research gaps, methods used, strengths and limitations of each studies are represented in Table 7.

Remarkably, the identified research gaps and the resultant findings reveals several similarities as well as inconsistencies. For example, several studies realized the need to estimate carbon sequestration potential and examine the economic viability of REDD + forest projects and community-based agroforestry. However, they differed in terms of the methods used and case study contexts, leading to certain inconsistencies in the findings. For instance, discussing in the context of agroforestry including community-based agroforestry projects, most studies found that such projects offer economic benefits (Table 7). However, the nature and extent of the benefits are subject to context-specific conditions including factors such as presence (or absence) of regulatory frameworks, inclusion (or exclusion) of community members, additional support, and alternative land use possibilities. For example, different types of Cocoa-agroforestry suggests different extent of sequestration and return potential in Indonesia (Santhyami and Suparti, 2023). Further, certain studies revealed positive economic benefits when the agroforestry projects are supported through CSR (Singh et al., 2024). Further discussing the context of REDD projects associated with either forestry or agroforestry and land use, most REDD + conservation efforts reveals challenges in achieving the intended benefit. Majority of the studies highlighted the poor returns from the establishment of REDD projects such as uncertain forest carbon stock estimates (Watson et al., 2013), low credit demand (Garcia et al., 2021), weak revenue generation (Martello et al., 2010), high opportunity costs (Butler et al., 2009), and governance conflicts (Boer, 2018).

Another critical research gap addressed include examining the governance issues surrounding VCM projects. While the studies reveal governance challenges and press for robust regulatory frameworks, the findings reveal different outcomes under different governance mechanism, for example, public governance (Boer, 2018), public-private (Moxey et al., 2021), and private and non-profit (Sharma et al., 2023). Though a greater involvement of private actors in project governance expedites the process, ethical concerns due to the increasing role of the private sector (Pettenella and Brotto, 2012), and disadvantages to the local community stakeholders due to the shrinking role of the state as mere facilitators of projects (Sharma et al., 2023) is highlighted.

Findings from the studies that examined research gap concerning land-use aspects, also reveals interesting yet contrasting outcomes that need further exploration. For instance, the conversion of disused ponds into mangroves offers both high carbon sequestration potential and strong financial returns (Cameron et al., 2019). However, in case of REDD projects, economically viable alternative land use options were preferred over forest conservation (Butler et al., 2009). Yet, there are studies that suggests pathways to promote sustainable land use for carbon sequestration, for example, blended finance models to fund such projects (Li and Martino, 2024). Several such findings, interesting yet reasonably inconsistent in its own right. and the resultant research gaps, which can be inferred from Table 7, requisites more detailed investigation.

Table 7
Comparative literature review table.

Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
Pinnschmidt et al. (2023)	Compared the economic and carbon performance of tropical mixed-species vs. monoculture plantations in Costa Rica under timber and carbon valuation schemes	Quantitative; Comparative simulation modelling; NPV Scenario analysis	Secondary data from past studies and govt. reports; Granularity: Fine; Temporal: provides 30 years future projection Spatial: two plantation sites in Costa Rica	Both monocultures and mixed-species plantations in Costa Rica offer carbon sequestration with economic benefits	Forest investors could benefit from establishing mixed-species plantations when reforestation plantation sites where species-specific site suitability is uncertain.	Detailed investigation on joint production of economic and carbon benefits in tropical monoculture plantation forestry	More real-world examples of tropical commercial mixed- species plantations necessary to validate the results and its applicability Analysis based on several simplified assumptions.
Garcia et al. (2021)	Assessed the legal, technical, and market viability of implementing REDD + projects on indigenous lands in the Brazilian Amazon, under recent Brazilian and UNFCCC policy developments.	Qualitative legal-policy analysis; Basic quantitative analysis using secondary data; interpretative synthesis	Secondary data from past studies, govt. reports, VERRA registry; Granularity: Coarse; Temporal: 2009–2020 Spatial: Indigenous lands in the Brazilian Amazon	Policy developments in Brazil requisite investigations if REDD + projects in Indigenous lands linked to VCMs could be useful in forest protection.	VCM linked REDD + projects face low credit demand, weak revenue generation, and high implementation costs; project viability rely on robust compliance frameworks.	Intersection of legal, technical and market requirements aspects on REDD + projects offers a comprehensive understanding from different yet interrelated dimensions.	Focusing only on projects in indigenous land and lack of a comparison of such projects with REDD + projects in other lands limits the inferences drawn
Santhiyami and Suparti (2023)	Compared the cocoa-based agroforestry systems (AFS) and monoculture cocoa farming in terms of carbon offset potential and financial returns under the voluntary carbon market in West Sumatra, Indonesia.	Scenario analysis; cost benefit modelling (NPV, BCR, and Payback period)	Primary data collected from interviews, field-based observations and previous studies; Granularity: Fine; Temporal: 20-year financial simulation; field data collected between Jan–May 2020 Spatial: 2 districts in Indonesia	Cocoa-based agroforestry systems financed through VCM seen as potential mitigation alternatives in Indonesia.	Cocoa–coconut agroforestry system offers the highest long-term return and sequestration potential compared to cocoa monoculture, cocoa-banana, cocoa-clove, and cocoa-timber).	The study Combines both ecological (carbon storage) and economic (NPV, BCR) evaluation under two schemes	Limited capacity among farmers for compliance with long-term monitoring and project rules restricts the expected research output and opportunity.
Naime et al. (2020)	Assessed the economic trade-offs in ecosystem services (ESs) during natural forest regeneration and evaluated their implications for Payment for Ecosystem Services (PES) schemes that incentivise tropical forest restoration.	Quantitative valuation, chrono-sequence study, ES modelling, and economic analysis under different land-use scenarios	Primary ecological measurements and socioeconomic survey data from farmer surveys, ecological plots, carbon price data, previous literature; Granularity: Fine; Temporal: Forest succession data in a chrono sequence design up to 45 years; socioeconomic survey conducted in 2014 Spatial: Pacific Coast of Jalisco, Mexico	Natural regeneration process of tropical dry forests of the Pacific Coast of Mexico offers opportunities to examine synergies and trade-offs across ESs and across stakeholders.	Young and intermediate secondary forests act as carbon sinks, while old-growth forests store more carbon. Landholders economically favour pasture and young forests due to short-term returns. VCM undervalues ecosystem services compared to the social cost of carbon.	Directly compares two economic valuation methods on the same ecosystem service, showing the relative advantages for policy making	Carbon stocks are likely to be underestimated in this study since only aboveground biomass was considered; can entail a hypothetical bias, limiting the generalizability of the results.
Boer (2018)	Analysed the evolving role of the Indonesian government and public bureaucracies in establishing a commercial regime for REDD + carbon trading, and how this	Case study analysis	Primary data through stakeholder interviews; Secondary data from govt. reports, policy documents; Granularity: Aggregated;	Indonesia was an early leader on REDD +, and set up the regulatory frameworks and monitoring systems to support projects that could trade credits in VCMs	Indonesia's REDD + regime diverges from neoliberal assumptions as the government agencies played a central role in shaping the institutional framework. Weak project outcomes	Focuses on the role of public agencies in creating enabling institutions, legislation, land-use planning, and MRV systems to support REDD + carbon markets.	Study focuses on institutional and governance processes and not on REDD + environmental or financial effectiveness.

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Table 7 (continued)

Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
	challenges the dominant neoliberal narrative in forest carbon markets.		<i>Temporal</i> : 2008–2013 <i>Spatial</i> : Subnational REDD + projects in Aceh and Central Kalimantan provinces in Indonesia		resulted from bureaucratic competition, fragmented authority, and governance conflicts.		
Singh et al. (2024)	Assessed the potential of carbon revenue in enhancing farmers' income through agroforestry-based carbon finance models in India.	Quantitative analysis, field sampling, data integration, carbon revenue estimation	Primary data collected from the study area; secondary data from previous studies; Granularity: Aggregated; <i>Temporal</i> : Based on current plantation age <i>Spatial</i> : Yamuna Nagar, India	Govt. of India see agroforestry as a potential source of carbon revenue to improve farmer's income	Agroforestry plantations can boost farmers' income by 40–50 % if linked with VCMs and corporate CSR funds. Carbon revenue can offer income stability like MSP in case of market volatility.	Integrates field-based carbon stock data with national-scale estimates offering both local and macro-level insights.	The estimation of additional income is based on not critically established assumption. The study lacks insights on potential trade-offs, which can be explored in future studies.
Zhang et al. (2024)	Assessed the economic, market, and policy constraints affecting afforestation as a climate mitigation tool in China.	Quantitative modelling (forest growth model, carbon pricing scenarios, Afforestation marginal abatement cost curves), and Scenario analysis	Secondary data from previous studies, govt. reports, national database, policy documents; Granularity: Aggregated; <i>Temporal</i> : 2020–2060 <i>Spatial</i> : Several provinces China	China, a leader in afforestation, and its evolving policy landscape for carbon markets offers opportunity to examine the implementation cost of climate mitigation through afforestation.	Implementation costs more relevant constraint to afforestation than opportunity costs. Aligning policy, raising carbon prices, and easing market access are essential to scale afforestation-based CO ₂ removal.	The study reveals more significant policy and market constraints beyond the physical and cost constraints. Use of NPV method over Maximum Potential Farm Revenue provides more accurate results.	Other factors (e.g., religion, personal preference, technology, level of information understanding) that affect afforestation not considered in detail, which may devoid a holistic understanding.
Cong et al. (2021)	Analysed the distribution and regional determinants of CCER (China Certified Emission Reduction) projects and assessed the economic rationale for the development of China's voluntary carbon market.	Panel data analysis	Secondary data from govt. reports, national policy documents, Registry database (CCER); Granularity: Fine; <i>Temporal</i> : 2010–2016 <i>Spatial</i> : China	2789 projects under China Certified Emission Reduction spread in 30 provinces studied to examine the regional determinants	VCM projects in China demonstrated economic sensitivity in project implementation. Projects are more likely in regions with lower per capita GDP, higher CO ₂ emissions and energy intensity, and better access to domestic loans.	Highlights the importance of regional economic factors in influencing emission reduction projects using a comprehensive list of projects spread across several provinces.	Uses registry data without assessing project outcomes or effectiveness. Stage of the projects and outcomes may lead to different implications.
Cameron et al. (2019)	Quantified GHG emissions and removal rates from degraded shrimp ponds and compared them with emissions from rehabilitated mangrove sites to assess their climate mitigation potential and cost-efficiency as potential carbon markets.	Carbon stock estimation in study site and economic return modelling	Primary data collected through field measurements; Secondary data from supplemented project records, govt. reports, previous studies; Granularity: Fine; <i>Temporal</i> : 2014–2016 <i>Spatial</i> : Coastal Sulawesi, Indonesia	Tropical mangrove ecosystems in coastal Indonesia offers high carbon storage and sequestration potential, yet capitalizing such potential remains unrealized.	Converting disused ponds into mangroves offers both high carbon sequestration potential and strong financial returns. Such rehabilitation outperforms many avoided deforestation projects in terms of net emissions reductions and investment appeal.	Combination of economic analysis and scenario analysis provides a very detailed assessment of mitigation benefits from mangrove rehabilitation.	Future studies may extended to projects which are certified under climate, community, and biodiversity standards.
Lupala et al. (2017)	Assessed the feasibility of implementing a community-managed carbon project under the REDD + framework, focusing on potential	Economic feasibility combined with scenario analysis	Primary data from household surveys and interviews; secondary data references from previous studies; Granularity: Fine;	Community based management of miombo woodlands in Tanzania is promoted for carbon credit project development.	Community-based management of Miombo woodlands is technically feasible and financially viable under VCM.	Outcomes of different economic feasibility analysis combined with scenario analysis provide a detailed understanding on viability of carbon projects	Institutional and policy context were not captured in this study.

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Table 7 (continued)

Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
	carbon benefits, economic viability, and community willingness.		<i>Temporal</i> : Data collected in 2012; carbon benefit projections over 20 years <i>Spatial</i> : Southern Highlands of Tanzania				
Martello et al. (2010)	Identified factors influencing leakage in REDD projects, and proposed intervention points to reduce leakage risks in tropical forests of developing countries.	Systems analysis using causal loop diagram	Secondary data from policy documents, previous studies, UNFCCC documents; Granularity: Coarse; <i>Temporal</i> : Documents available till 2010 <i>Spatial</i> : Developing countries with tropical forests	Tropical forests under REDD in developing countries with leakage challenges	Constraints related to economies of scale, political instability and lack of institutional capacity hamper success of REDD projects. Cross-border leakage may have negative consequences for local forest conservation and livelihood outcomes.	Use of systems analysis offers a holistic view of forest-based carbon projects	Lack quantitative data or analysis limits the generalizability of the outcomes; Systems Dynamic Model may be explored in future
Lasco et al. (2010)	Assessed the climate mitigation potential of Community-Based Forest Management (CBFM) and whether local participants can benefit from global carbon markets (CDM/ REDD+).	Quantitative estimates of carbon stocks and sequestration	Secondary data included biomass estimates and data from the govt. reports, UBFCDC database; Granularity: Coarse; <i>Temporal</i> : In 2010 <i>Spatial</i> : Philippines	CBFM agroforestry projects in Philippines linked to carbon markets for climate mitigation	Transaction cost in CDM markets are potential constraints; VCMs could be an alternative due to its flexibility	A detailed macro-level overview on the potential of agroforestry projects for carbon mitigation in Philippines	Carbon sequestration rates of trees, and economic analysis of agroforestry carbon Projects specific to the Philippines.
Aerni (2016)	Examined the field-level experiences with PES projects, alongside assessing the role of the private sector and innovation in scaling PES.	Comparative case study approach	Primary data through interviews from stakeholders; Secondary data from policy, and project documents; Granularity: Fine; <i>Temporal</i> : Early 2000s-2012 <i>Spatial</i> : Kenya	Kenya Agricultural Carbon Project and other Payment for Ecosystem Services projects designed to benefit farmers from VCMs	The presence of the private sector is crucial in making use of Payment for Ecosystem Service projects as a vehicle to develop VCMs	Diverse real-world case studies offer rich and in-depth insights into specific situations.	Qualitative case study approach lacking quantitative analysis limits the generalizability of the findings
Pettenella and Brotto (2012)	Identified the governance features that contribute to the success or failure of REDD + projects.	Exploratory case study approach, social network analysis, document analysis	Primary data collected from field visits, semi-structured interviews; Secondary data from Project design documents (PDDs); Granularity: Fine; <i>Temporal</i> : In 2011 <i>Spatial</i> : Peru	Successful REDD + projects in Amazon offers governance lessons for VCM	Strong Forest governance, especially land use planning, certification, transparency, and accountability, is essential for REDD + success. Involvement of the private sector may expedite processes but may raise ethical concerns.	The study attempts to develop a practical, indicator-based framework for assessing governance quality in REDD + projects at the project level.	The use of only two primary case studies (one successful, one failing) limits the robustness of the comparative analysis.
Grace et al. (2010)	Presented results from a pilot VCM project in Mozambique focused on tree planting and woodland protection, and its associated economic and social impacts.	Field plots, Remote sensing, Baseline modelling	Primary data collected through interviews; Secondary data sourced from World Bank database, Plan Vivo PDD, SPOT-4 and ALOS; Granularity: Fine; <i>Temporal</i> : 2003–2009	Mozambican pilot project linked to VCM could be used to motivate communities to avoid deforestation and plant trees.	The project demonstrates that community-based agroforestry and forest protection under the Plan Vivo system yields co-benefits. Avoided deforestation yielded more carbon credits than tree planting.	Provides empirical, field-based evidence from a long-term pilot implementation on aspects such as biomass measurements, remote sensing, and socio-economic surveys.	High heterogeneity in the Miombo landscape necessitated parcel-specific biomass estimation, making scaling difficult without considerable effort.

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Table 7 (continued)

Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
Galgani et al. (2014)	Assessed the climate and economic benefits of composting, anaerobic digestion, and biochar production for organic waste recycling and evaluated their viability under VCMs.	Life Cycle Assessment, Economic Assessment, Scenario Analysis	<i>Spatial:</i> Sofala, Mozambique Primary data collected through interviews and observations, field trials; Secondary data sourced from pilot data, online databases; <i>Granularity:</i> Fine; <i>Temporal:</i> Modelled one-year operational cycle <i>Spatial:</i> Tamale, Northern Ghana	Organic waste recycled in Tamale and the surrounding peri-urban areas within 50 km	Biochar offers the greatest climate mitigation potential and could become viable under VCMs while Composting and Anaerobic Digestion would require much higher carbon prices to break even.	It contributes to an under-researched area by providing a Ghanaian case study that evaluates the potential of carbon markets to support sustainable waste management technologies.	The economic assessment can include costs associated with carbon project accreditation and monitoring, assuming these are borne by external consultants in the future.
Kalaba et al. (2013)	Evaluated aboveground carbon storage and floristic structure in fallows from charcoal and agricultural use, and assessed implications for carbon management in Miombo woodlands.	Biodiversity analysis, Regression analysis	Primary data regarding carbon storage; <i>Granularity:</i> Fine; <i>Temporal:</i> In 2012 <i>Spatial:</i> Miombo woodlands, Zambia	Wet Miombo woodlands in Copperbelt Charcoal and agriculture fallows in Zambia seen as potential carbon-based ecosystems services to restore carbon and biodiversity	Post-abandonment regrowth after charcoal and agriculture restores carbon stock to levels comparable with mature woodlands in approx. 20 years. Natural regenerations can be incentivized under carbon markets. Charcoal and slash-and-burn systems have restoration potential if managed sustainably.	Provides empirical evidence on C storage and how C and biodiversity change after disturbances from the main drivers of forest loss in tropical drylands.	Although 20 years is sufficient for the forest structure of regrowing Miombo to resemble mature woodlands, this time is not sufficient for the floristic composition to recover.
To et al. (2017)	Examined how REDD + interacts with local land tenure systems and agrarian change in shaping property rights and land-use decisions.	Ethnographic and qualitative research	Primary data from surveys, participant observations; Secondary data from project documents, govt. records; <i>Granularity:</i> Fine; <i>Temporal:</i> 2014 <i>Spatial:</i> Central Highlands of Vietnam	REDD + influencing property rights and land use in central highlands of Vietnam	REDD+ was shaped by local institutions and used by communities more for local objectives, such as securing land tenure and expanding agricultural livelihoods, than for carbon sequestration, undermining REDD + goals.	Conceptually frames REDD + as a process of co-production, offering a nuanced understanding of how transnational environmental governance interacts with local property rights and livelihood.	Future studies may consider economic aspects such as opportunity-cost of land use to negotiate land rights related to nature-based carbon projects
Paiva et al. (2014)	Assessed the co-benefits of reforestation projects in the VCM using the Monte Pascoal-Pau Brazil Ecological Corridor Project (MPPBCEP) in Brazil as a case study.	Case analysis, document analysis,	Primary data collected through interviews, and field visits; Secondary data from PDDs, institutional reports; <i>Granularity:</i> Fine; <i>Temporal:</i> In 2012 <i>Spatial:</i> Bahia, Brazil	The evolving reforestation projects linked to VCMs in Brazil expected to offer several co-benefits.	Forest carbon projects offer significant environmental and social co-benefits, including participatory governance. The absence of a robust institutional framework causes legal and policy uncertainty, threatening long-term sustainability. Local livelihoods heavily depend on fisheries supported by mangroves that store moderate carbon, mostly in soils, yet represent an unrealized financial asset.	A holistic assessment of sustainable development benefits using a UNFCCC-aligned co-benefits framework.	Land tenure complexities and hesitance from landowners limit broader adoption as well as restrict the research horizon.
Benson et al. (2017)	Assessed mangrove carbon stocks and land cover change in Helodrano Fagnemotse and discussed implications for local	Field sampling, remote sensing analysis	Primary data from field carbon stocks measurement, Landsat satellite images; <i>Granularity:</i> Aggregated;	Mangroves gaining attention for carbon storage in southwest Madagascar		Integrates remote sensing with carbon measurements, offering both spatial and biophysical data to understand landscape-level carbon storage dynamics.	Drivers behind cover dynamics (e.g., human pressures, climate factors), which can be underlying causes of degradation, can

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	livelihoods and carbon finance.		<i>Temporal</i> : 2002–2014 <i>Spatial</i> : Southwest Madagascar		Carbon project methodologies need to be adapted to wetland ecosystems for better applicability.		be assessed in future studies.
Chayaporn et al. (2021)	Assessed the overall carbon storage and economic potential of a teak plantation under forest management from planting to final felling.	Allometric equation modelling, Logistic growth modelling	Primary data from field survey, interviews; Secondary data from past studies; Granularity: Fine; <i>Temporal</i> : Carbon modelled over 105 years (3 management cycles of 35 years each) <i>Spatial</i> : Western Thailand	Teak Plantation under forest management in Thailand investigated for carbon storage and economic potential.	Teak plantations managed through periodic thinning and final felling can significantly contribute to carbon sequestration. Recommends technology transfer and carbon-finance policies to support the long-term preservation of carbon.	It presents a comprehensive and integrated assessment of total carbon storage across standing biomass and HWP's (Harvested Wood Products).	Results are specific to Thong Pha Phum plantation and depend on thinning intensity, species, and local market demand, limiting generalizability.
Kongmeesup and Boonyanuphap (2019)	Assessed carbon stock and sequestration in teak plantations aged 21–36 years and estimated the economic value of carbon offsets through the VCM.	Allometric modelling, Walkley & Black method, Empirical carbon stock and offset value analysis	Primary data collected from field measurements; Secondary data from previous studies; Granularity: Fine; <i>Temporal</i> : 2014–2019 <i>Spatial</i> : Lower Northern, Thailand	Teak Plantation under forest management in Thailand investigated for carbon storage and economic potential	The evaluation of income from carbon offsets in teak plantations highly correlates with the age of the teak. The total carbon stock of teak aged between 21 and 24 years had the lowest total carbon stock, whereas the highest total carbon stock was found in teak plantation aged between 35 and 36 years.	Successfully quantifies carbon sequestration potential across a full range of teak plantation and translates it into this into potential VCM revenue.	The carbon offset values are based on simulated 5-year contracts using fixed assumptions, which may differ from actual market conditions.
Bomfim et al. (2022)	Addressed the lack of integration between carbon sequestration and sustainable forest management (SFM) in tropical forests by proposing a carbon-inclusive SFM framework (SFM + C) supported by VCM.	Conceptual framework development, Quantitative modelling	Secondary data from past studies, carbon indices data, govt. reports; Granularity: Coarse; <i>Temporal</i> : Simulations projected over a 40-year harvest cycle <i>Spatial</i> : Brazilian Amazon	Tropical forests in Brazilian Amazon could provide crucial natural climate solutions and opportunities in the evolving VCM	Financial viability improves when combining carbon credit revenue with timber income under voluntary carbon markets. SFM + C provides a viable climate mitigation strategy while maintaining sustainable timber yields.	It combines policy, ecology, and market finance to propose a measurable strategy for enhancing tropical forest carbon storage.	The study uses secondary data and assumptions rather than field measurements, emphasizing the need for primary data validation of species/site-specific SFM + C strategies.
Lock (2024)	Conducts a critical political-economic analysis of how surplus value is produced and captured across value chains in VCMs by exposing the unequal value distribution, and the roles of narratives and networks in creating legitimacy and justifying high prices for carbon credits.	Ethnographic fieldwork, discourse analysis, social network analysis	Primary data collected from stakeholders; Secondary data from PDDs, certification reports; Granularity: Fine; <i>Temporal</i> : Fieldwork done from 2017 to 2019, market document analysis from 2015 to 2023 <i>Spatial</i> : San Martin, Peru	Value creation and capture in VCM in context of REDD + projects in Peru	VCMs generate surplus value through narratives of innovation, permanence, and co-benefits, which drive up credit prices. Brokers, auditors, and financial intermediaries capture much of this surplus, while growers receive a small share.	It offers a novel critical lens, blending ethnography, network, and discourse analysis.	Broader comparative work is needed to assess how surplus value extraction operates in different contexts and commodity systems.
Butler et al. (2009)	Examined whether REDD schemes-currently limited	Economic modelling; scenario analysis; NPV	Secondary data from institutional databases,	Tropical forests of Indonesia that can be	REDD schemes under voluntary market prices are	A rigorous economic comparison of REDD and oil	It excludes payments for environmental services

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Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
	to voluntary markets-can offer a financially viable alternative to palm oil cultivation, a major driver of tropical deforestation.	estimations; sensitivity Analysis	exchange data, previous studies; Granularity: Coarse; Temporal: 30-year model projection (2009–2039) Spatial: Indonesia	converted to REDD + project or alternative land use such as palm oil cultivation.	significantly less profitable than oil palm cultivation. High global demand for palm oil and increasing prices make oil palm an economically attractive alternative. Without policy reform to include REDD in compliance carbon markets, its uptake will remain low.	palm development, highlighting how market structure and policy design fundamentally shape land-use decisions.	(PES) beyond carbon, including forest-derived goods and services that benefit local and regional economies.
Cariappa et al. (2024)	Assessed conservation agriculture (CA) practices for their eligibility in generating carbon credits under VCM protocols.	Carbon sequestration potential modelling, Systematic field review, Mapping of farm-level practices	Primary data from field observations; Secondary data from VERRA documentation, previous studies, carbon price indices; Granularity: Coarse; Temporal: Multi-year carbon rate models; likely analysis period 2010–2020 Spatial: Punjab and Bihar, India	Conservation agricultural practices seen as potential ways to generate carbon credits in India.	Conservation agriculture practices can qualify for voluntary carbon credits when aligned with approved methodologies. Eligibility is higher where CA includes residue retention and reduced tillage, emphasizing the importance of practice bundling.	It establishes a practical eligibility assessment bridging policy and farmer-level implementation.	Uncertainties remain in measured sequestration rates under diverse local conditions, and additional field validation is necessary to confirm eligibility across all agro-climatic zones.
Kent and Hannay (2020)	Explored how the concept of “carbon” is explained and understood in community-based sequestration projects and the role of communication in creating local carbon knowledge.	Thematic content analysis	Primary data collected through interviews with selected stakeholders; Granularity: Fine; Temporal: In 2019 Spatial: Multinational (Ghana, Tanzania, Burkina Faso, Kenya, Brazil, Colombia, Guatemala, and Indonesia)	Formation of local carbon knowledge is central to the meaningful participation of communities in the land-based carbon projects.	Facilitators struggle to explain the intangible and abstract nature of “carbon,” often translating it through metaphors, local knowledge, and visible analogies. The communication strategy must be co-designed with community members.	Provides critical insights into the local formation of carbon knowledge.	The study does not capture the voices of community members themselves, focusing only on facilitators.
Somprasong (2024)	Employed remote sensing and GIS-based spatial techniques to estimate both carbon sequestration potential and its economic value in post-mining landscapes.	Integrated spatial analysis, Relative Standard Deviation analysis, economic valuation	Secondary data collected from satellite data, previous studies, institutional databases; Granularity: Coarse; Temporal: Economic estimates projected till 2030 Spatial: Thailand	Reclamation of mining areas and reforestation as a carbon sequestration strategy in Thailand.	Northern Thailand has the highest carbon sequestration potential due to extensive vegetated areas and closure-stage mines while economic value is highest in the Northeast. Remote sensing and NDVI-based models offer an effective framework for national-scale carbon stock estimation in reforested mine.	Quantifies both the biophysical and economic potential of mine-reforestation for carbon sequestration validation.	Even supposing that the precision of the estimation can be statistically acceptable, there is some evidence of the high uncertainty of estimations.
Duker et al. (2019)	Analysed how REDD + projects are implemented and why they fail to meet the livelihood needs of	Extended case study approach, thematic analysis	Primary data from semi-structured interviews, focused group discussions,	REDD + projects often fail taking into account smallholder agricultural	Both projects suffered from ad hoc, donor-driven financing, and lacked long-term financial security.	Contributes detailed empirical evidence from the REDD + pilot project	Statistically representative conclusions were a challenge and could not be reported due to the sample

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Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
	smallholder farmers, leading to distrust, marginalization, and carbon leakage.		validation workshops; Secondary data from project documents and academic literature; Granularity: Fine; <i>Temporal</i> : Fieldwork conducted between November 2014 and September 2016 <i>Spatial</i> : Two sites in Indonesia and Ethiopia	interests in project design and implementation.	Without integrating smallholder agriculture and livelihood needs, REDD + will fail to achieve real emission reductions and may instead trigger carbon leakage and livelihood insecurity.	implementation in two different contexts.	not being adequate for varied geographies.
Indrajaya et al. (2024)	Evaluated the cost-effectiveness of the Verified Carbon Standard (VCS) remuneration scheme compared to a dynamic “current carbon accounting” method based on actual yearly carbon storage.	Economic modelling, Sensitivity Analysis, NPV analysis	Secondary data sourced from plantation database, carbon stock metrics, govt. reports, voluntary market indices; Granularity: Coarse; <i>Temporal</i> : 30-year rotation baseline modelling (extendable) <i>Spatial</i> : Indonesia	Different remuneration scheme for forest carbon management may have different outcomes.	VCS is less cost-effective than current carbon remuneration methods, as it doesn't align well with the actual yearly variation in carbon storage. Current VCS prices are too low to incentivise longer rotation lengths for <i>A. mangium</i> plantations in Indonesia.	Compares the costs of carbon sequestration under the VCS rule with a current carbon approach, highlighting the influence of accounting design on forest management incentives.	Focuses solely on carbon and timber values; belowground carbon could also be accounted for.
Cariappa et al. (2024)	Assessed the inclusivity, additionality, and permanence of carbon farming projects in India's VCM.	Two-stage probit modelling	Primary data collected through surveys; Secondary data from Verra documents, previous studies; Granularity: Fine; <i>Temporal</i> : In 2023 <i>Spatial</i> : Haryana (North India) and Madhya Pradesh (Central India)	Emerging carbon farming projects across the Indian states offers opportunities to identify early-stage challenges	Participants skewed toward large landholders and non-marginalized castes out of which 99 % of farmers received no monetary benefit. Carbon-focused firms (Carbon Core) were more effective than diversified companies in spreading regenerative practices.	Assesses carbon farming in India from a social equity and carbon market integrity perspective, integrating farmer behaviour, practice timelines, and carbon credit methodologies.	Study recommends longitudinal tracking of marginal farmer experiences over multiple carbon credit issuance cycles, which could enhance causal inference on permanence.
Moreno et al. (2019)	Addressed the lack of integrated forest management models for native, slow-growing forests that incorporate carbon storage into optimal harvest decision-making.	Bioeconomic modelling, Scenario analysis, Allometric modelling	Primary data collected through forest sampling; Secondary data from govt. reports, previous studies, online databases; Granularity: Fine; <i>Temporal</i> : Economic simulations modelled over a 100-year horizon <i>Spatial</i> : Malleco National Reserve, Chile	Investment in native forest management under subsidy framework in Chile	With carbon and forest management, optimal harvest cycle increases from 59 to 62 to 62–64 years. It improves NPV and IRR as well significantly compared to timber-only. Inclusion of carbon credits could make native forest management economically competitive.	The study provides attractive results integrating tangible assets and environmental services to the forest multifunctionality target.	The sampling intensity should have been higher to reduce estimation error due to the heterogeneity of these multi-aged forests but field inventory difficulties (forest cover, slope and penetrability) and limited budget constrained this.
Cuesta et al. (2023)	Provided pixel-level AGC mapping with robust uncertainty estimates for the Choco Andino region to measure above-ground carbon stocks.	Integrated modelling, remote sensing, systematic elevation-stratified RF modelling	Data type: Primary data collected from local plots; Secondary data from regional databases, satellite imagery; Granularity: Fine; <i>Temporal</i> : 2018–2021	Carbon compensation projects could significantly aid in preserving Tropical Mountain Forest in the Ecuadorian Andes	AGC density strongly varies with elevation, with the highest stocks in the piedmont and lower mountain zones.	The use of high-resolution maps displaying carbon stocks on a per-pixel level with high accuracy demonstrates a cost-effective, scalable method for spatially explicit carbon mapping.	Despite statistically acceptable precision, unavoidable uncertainties persist due to remote sensing limitations and terrain complexity.

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Study (Author, Year)	Research gap fulfilled by the study	Methods	Data Type & sources; Granularity; Coverage (represented in chronological order)	Study/Case study context	Key Findings	Strength	Weaknesses/Future research direction or opportunity
Moxey et al., 2021	Addressed the gap in understanding the specific barriers faced by land managers and practitioners when trying to implement the UK Peatland Code within a blended finance model, combining public and private funding.	Case study analysis	<i>Spatial:</i> Ecuadorian Andes Primary data from field observations; Secondary data from policy documents, programme documents, previous studies; <i>Granularity:</i> Medium; <i>Temporal:</i> 2015–2020 <i>Spatial:</i> United Kingdom	Implementation of the UK Peatland code (blended green finance) to reduce carbon emission	Uptake of the Peatland Code has been slow despite increased VCM interest. Improved awareness, carbon credit payments, flexible schemes, and coordination between public-private initiatives can support peatland restoration.	As a case study of the barriers and opportunities to blended finance, the observations offered will have relevance to other applications elsewhere.	Focused on a specific application in one country and hence limits generalizability.
Nonini and Fiala (2021)	Estimated total carbon storage (TCS) of forest biomass at the stand level by including AGB, BGB, deadwood, and litter for VCM relevance.	Spreadsheet-based modelling, scenario analysis	Secondary data collected from Forest Management Plan data, govt. data, previous literature; <i>Granularity:</i> Fine; <i>Temporal:</i> Projection until 2029 under various scenarios <i>Spatial:</i> Northern Italy	Estimation of carbon storage of forests essential to develop local VCM	Multifunctional Forest management can slightly enhance sequestration. By 2029, converting coppices to high forests could enhance total carbon storage, offering a modest improvement over business-as-usual.	The study presents a simplified model specifically developed to estimate total carbon storage by using site-specific primary data.	Uses simplified assumptions and ignores dynamic effects.
Ezquerro et al. (2024)	Proposed methodology for evaluating improved forest management (IFM) in strategic planning for forests with long-term management history; to assess trade-offs with biodiversity; and to manage long-term carbon capture and timber harvests.	Multi-criteria decision analysis (MCDA), scenario comparisons using simulations	Secondary data collected from historical forest inventory, economic data, carbon pricing assumptions; <i>Granularity:</i> Coarse; <i>Temporal:</i> 10-year planning horizon <i>Spatial:</i> Valsain Forests in Spain	Trade-offs and synergies exist between carbon capture and other ES, which needs consideration under integrated Forest Management for favourable outcomes.	IFM integration allows for increased carbon capture in the upcoming 10-year period. Trade-offs between carbon capture and biodiversity are evident, but compromise solutions are achievable through MCDA.	Underscores the importance of policy support in establishing clear standards for voluntary markets and promoting the accessibility of carbon offsets.	Results may not generalise to other forest types or regions due to varied geography.
Meitern (2024)	Investigated whether carbon credits could serve as a viable financing mechanism for local energy communities, supporting sustainable energy transitions and climate mitigation.	Qualitative assessment, policy analysis	Primary data through project survey, Secondary data from previous studies, policy reports, business cases; <i>Granularity:</i> Coarse; <i>Temporal:</i> Projects selected for review spanning from 2018 to 2025 <i>Spatial:</i> Several countries in Europe	Innovative local energy communities in Europe, supported by carbon markets, have significant carbon emissions reduction potential	The ten innovative projects respond to carbon credits' main quality criteria; thus, they can significantly lower carbon emissions, operate in addition to existing practices, employ robust measurement mechanisms, and align with Sustainable Development Goals (SDGs).	The study underscores the importance of policy support in establishing clear standards for voluntary markets and promoting the accessibility of carbon offsets.	The results of this analysis should be considered with caution due to some data limitations. For e.g., Horizon projects typically lack rigorous audits for their carbon reduction claims.
Sharma et al. (2023)	Examined the role of the state in the rescaling of forestry governance to private and non-profit sectors, and how this affects access, power, and equity in land use.	Case study analysis, Document review	Primary data collected through interviews; Secondary data from policy documents, statistical reports; <i>Granularity:</i> Fine; <i>Temporal:</i> 1990–2020 <i>Spatial:</i> Scotland, UK	Scottish Government's policies increasingly rescale forest and landscape governance to private and non-profit actors	Market-based restoration has made land access increasingly unaffordable for communities, favouring green investors. Forestry policies have shifted from state-led	The study has been able to shed light on power asymmetries both between and within global and local levels.	The study did not interview private or community landowners directly, as they can be highly heterogeneous within the Scottish context.

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Ranjan (2024)	Addressed the limited exploration of how greenwashing risks affect firm participation in VCMs. It also highlights the gap in modelling interlinked formal and voluntary markets and their dynamic interaction with community conservation norms.	Simulation-based modelling, scenario analysis	Primary data collected through field observations; Secondary data from previous studies, national datasets, online databases Granularity: Fine; Temporal: Simulation is conducted over a 50-year time horizon Spatial: India	The case of a large Indian steel firm issuing green bonds to fund afforestation project in Himalayan Forest communities	implementation to state-facilitated regulation, with private actors now taking lead roles in afforestation and woodland creation. Linking formal and VCM can generate cost-effective carbon reduction options for firms while positively reinforcing forest conservation. Forest-dependent communities gain income by conserving forests, improving forest health.	By developing a model of firms' decision-making processes for mitigating carbon emissions through various options, this study sheds light on the trade-offs involved.	Further research is needed to evaluate the effectiveness of community norms in reducing illegal forest harvesting and the success of VCMs. Additionally, this study did not consider the potential impact of climate change on community conservation norms.
Zheng and Zhang (2023)	Investigated the regional spillover effects of China's voluntary carbon market under the ETS, especially its unintended causal effect on urban greening.	Quasi-experimental approach using the Difference-in-Differences (DiD) estimation technique, Event study regressions and robustness checks	Secondary data sourced from MODIS (NASA), Emission exchange data, national datasets, govt. reports, published yearbooks for socio-economic data; Granularity: Fine; Temporal: 2010–2019 Spatial: Guangdong Province, China	The pilot policy of China's regional offset mechanism offers opportunities to examine the regional ecological benefits of VCM projects.	Positive spillover effects are not significant in high-emission cities or where many firms already participate in the carbon market. Policymaking should integrate carbon market mechanisms with forestry, urban planning, and poverty reduction strategies.	The study conducts a rigorous quasi-experimental identification using DiD and event study designs to isolate the causal impact of a voluntary carbon market mechanism (PHCER) on urban greening.	There are potential limits in scaling the pilot experience until the experience is effectively replicated.
Birchall et al. (2018)	Explored development of the VCM in New Zealand during its early emergence, addressing the gap in the academic literature concerned with lack of studies conducting in-depth studies surrounding new institutions.	Case study analysis, thematic analysis	Primary data through interviews; Granularity: Fine; Temporal: 2010–2011 Spatial: New Zealand	The emerging phase of VCM in New Zealand with specific cases drawn from the wine, taxi, and carbon services industries.	Early market development included efforts to promote integrity through institutional infrastructure (registry, certification programs). and hesitance to participate due to reputational risks from scrutiny over climate claims. Over time, carbon offset costs decreased and climate-friendly technologies (like hybrid vehicles) gained prominence in the market.	The study conducts a critical investigation at the organizational and field levels to better understand the institutions that developed in the early stages of New Zealand's VCM.	While this study provides insight into the early stages of the VCM, the state of the market in New Zealand has indeed changed since the study occurred. Thus, a follow-up study is in development to provide a longitudinal dimension.
Almeida et al. (2014)	Assessed the carbon footprint of development aid organizations, particularly within the North-South cooperation framework.	Life Cycle Assessment, Sensitivity Analysis	Primary data collected from questionnaire responses; secondary data from Institutional reports, online databases; Granularity: Coarse; Coverage: Temporal: Carbon footprint data from 2009	Development organizations across Belgium and Germany that advocate sustainability, whose operational environmental performance remains unassessed.	Electricity and heating, staff mobility significantly contribute to organizational footprints, with air travel contributing approx. 40 %. Voluntary carbon markets or in-project offsets can partially compensate for emissions.	This is the first empirical effort to quantify and compare the carbon footprints of development aid institutions.	Limited data availability, lack of standardized reporting, and exclusion of Southern operations limit generalizability.

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Dwivedi et al. (2009)	Integrated life cycle emissions with economic modelling under Chicago Climate Exchange (CCX) guidelines to evaluate the net benefits of carbon payments in slash pine management.	Integrated life cycle analysis, Sensitivity Analysis, Faustmann economic model	<i>Spatial:</i> Belgium (4 development agencies), Germany (GIZ used for comparison) Primary data from expert interviews; Secondary data from past studies, published databases, exchange data; <i>Granularity:</i> Fine; <i>Temporal:</i> Simulated plantation rotation cycles over 30 years <i>Spatial:</i> Southern part of USA	VCMS provides additional income opportunity to nonindustrial private forest landowners in USA through carbon sequestered forest biomass.	Carbon payments significantly increase land profitability under all scenarios and have minimal effect on optimal rotation age, implying low distortion risk to timber supply.	This study takes a holistic view by incorporating LCA and an advanced forest growth and yield model with forest economics.	Results may vary with the adopted silvicultural practices and the ecological characteristics of the forestland.
Birchall et al. (2018)	Examined how organizations interact with New Zealand's VCM via CarboNZero and CEMARS certifications, and assess emission reduction outcomes.	Qualitative document analysis, Quantitative tabulation of emissions trends	Secondary data extracted from disclosure documents, certification data, organisational data; <i>Granularity:</i> Coarse; <i>Temporal:</i> 2005–2011 <i>Spatial:</i> New Zealand	Climate mitigation related outcomes of organizations associated with VCM certifications in New Zealand.	CEMARS certification schemes offer lower-cost offsetting, appealing to emission-intensive organizations. External factors, especially from large emission-intensive players and environmental variability, significantly influence organisational outcomes and can mask genuine reduction efforts by smaller actors.	The study documents the entire range of consumption, production, and services related to the carbon offsetting industry in New Zealand.	Document analysis was limited to total organisational emissions and did not include any formal discourse analysis of the text in the disclosures.
Thompson and Hansen (2012)	Assessed large industrial forestland owners' intentions to manage forests for carbon offset production using the Theory of Planned Behaviour (TPB), and analysed the influence of policy expectations, ROI pressures, and beliefs about climate mitigation.	Extended Theory of Planned Behaviour (TPB), descriptive statistics, regression analysis, thematic analysis	Primary data from web-based questionnaire, organisational data; <i>Granularity:</i> Fine; <i>Temporal:</i> 2010–2011 <i>Spatial:</i> United States	VCMS and the national emission reduction legislation in the US make provisions for substantial carbon offsets from domestic forestry projects.	A passive 'wait-and-see' strategy dominates due to uncertainty in policy and low carbon prices. Firms vary in readiness depending on their structure, investment horizons, and exposure to institutional pressures.	It empirically examines the behavioural intentions of large forestland owners in the US regarding carbon markets using an extended TPB model offering qualitative insights.	The target population and respondent pool were small and a deeper understanding could have been attained with personal interviews rather than a web-based survey.
Li and Martino (2024)	Assessed the economic feasibility of VCMS in Scottish saltmarshes by comparing carbon stock gains from reinstated tidal flows versus land under agriculture and to provide market design suggestions.	Cost-benefit analysis, scenario analysis, sensitivity analysis	Secondary data from previous studies, project reports, market data, assumptions based on grey literature, govt. frameworks; <i>Granularity:</i> Coarse; <i>Temporal:</i> 40-year time horizon for economic evaluation	Temperate saltmarsh ecosystems in coastal Scotland under different land use scenarios.	Under current market prices, saltmarsh projects are not economically feasible unless carbon prices increase or additional funding mechanisms are included. Saltmarsh carbon projects in the UK will likely require hybrid finance models combining VCM revenue and public funding.	Conducts a comprehensive cost-benefit analysis of saltmarsh land-use scenarios providing a practical framework to evaluate feasibility under different price and management assumptions.	The lack of site-specific data on UK saltmarsh carbon sequestration rates, restoration costs, and VCM transaction costs limits the accuracy of the model. The study does not account for legal, administrative, or transaction barriers that may arise during actual VCM project implementation.

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Cammarata et al. (2024)	Analysed farmers' willingness to participate in soil carbon sequestration schemes under the VCM using an extended Theory of Planned Behaviour (TPB) framework.	Extended Theory of Planned Behaviour (ETPB) framework, PLS-SEM	<i>Spatial:</i> Scottish saltmarshes Primary data collected through questionnaire, survey; Granularity: Fine; <i>Temporal:</i> In 2024 <i>Spatial:</i> Sicily, Italy	Participation of European farmers in VCM	Farmers' attitudes, perceived behavioural control, and knowledge of VCM positively influence their intention to participate. Extension and technical assistance should be prioritized to improve farmer participation in soil carbon projects.	Detailed behavioural insight into farmers' decision-making, offering clear implications for capacity-building interventions.	The study suggests the need for targeted extension services and technical assistance, suggesting current informational limitations in farmer engagement.
Plieninger (2011)	Assessed whether agroforestry systems can sequester carbon and qualify as carbon offset projects, integrating landscape conservation with climate mitigation.	Case study analysis, spatial analysis, qualitative assessment	Primary data collected through digitized aerial photographs; Secondary data from previous studies, survey reports; Granularity: Fine; <i>Temporal:</i> Comparative analysis over 44 years (1964–2008) <i>Spatial:</i> Germany	Agricultural landscape and forestry in Germany seen as an effective carbon store and deliver important ecosystem services.	Six agroforest classes provide both carbon sequestration and co-benefits as along with meeting additionality, permanence, and low leakage risk criteria. The integration of scattered tree systems could serve as a low-cost, multifunctional land use compatible with carbon market participation.	Assesses the potential of agroforests in Central European agricultural landscapes to sequester carbon and participate in carbon markets, uniquely combining landscape ecology and climate policy.	Lack of direct carbon measurement and the absence of empirical carbon quantification for specific agroforestry systems remains a limitation.
McHale et al. (2007)	Assessed whether urban tree-planting projects can qualify as cost-effective carbon offset activities and identified key cost-effectiveness drivers.	Sensitivity analysis, Assessment of the impact of variables, Cost-benefit analysis, Implementation of urban forestry model	Secondary from previous literature, aerial reports, govt. database, assumptions based on past studies; Granularity: Fine; <i>Temporal:</i> 40-year simulation model <i>Spatial:</i> Colorado, USA	Potential of connecting urban tea planting to VCMs	Large-stature trees in long growing seasons and low-cost maintenance contexts can be cost-effective for carbon markets. Carbon assimilation rate is the most influential variable.	Alongside examining cost-effectiveness, it applies model sensitivity analysis as well to determine what variables most influence cost-effectiveness.	Relies on simulations rather than empirical monitoring; geographically limited.
Stivins et al. (2024)	Addressed the lack of practicable, field-based guidance on implementing paludiculture and voluntary carbon certification on formerly extracted peatlands in temperate regions.	Geological survey, baseline assessment, field mapping, hydrogeological monitoring.	Primary data from field measurements, peat hydrochemistry, vegetation mapping; Granularity: Fine; <i>Temporal:</i> 2023-24 <i>Spatial:</i> Latvia	VCMs open horizons for private companies, public institutions, and individuals developing carbon removal projects in peatlands to reduce overall carbon footprint	Demonstrates that private actors can establish carbon certification-ready baselines on extracted peatlands.	The study applies site-specific, measurement-based approaches to assess carbon stocks and vegetation heterogeneity.	It presents only baseline stock and initial chemical conditions, without integrating longitudinal flux measurements.
Karhunmaa et al. (2023)	Explored Finnish citizens' perceptions, understanding, and trust in voluntary carbon offsetting as a form of individual climate action.	Content analysis, regression analysis	Primary data collected through an online questionnaire; Granularity: Fine; <i>Temporal:</i> In 2021 <i>Spatial:</i> Finland	Carbon offsetting remains unsettled and ambiguously understood by citizens	Citizens show ambivalence-willing to act on climate but distrustful of voluntary carbon offsets. No strong or unqualified support; significant uncertainty and skepticism toward the sector.	The study builds a nuanced picture of citizen perceptions contributing to a growing research body on voluntary carbon offsetting.	Survey did not ask for definitions, making interpretations of responses uncertain.
Dufour et al. (2024)	Analysed risks of co-financing BECCS through both state support and VCM credits, and proposed	Case study analysis	Secondary data from Swedish BECCS policy framework, market agreements, and literature sources;	Financing of bioenergy with carbon capture and storage (BECCS) in Sweden	Found that state subsidies for BECCS under current policy would lower VCM prices, which may reduce decarbonization incentives	Provides a policy-relevant framework and clear recommendations for safeguarding integrity in co-financed BECCS projects,	Focus is on theoretical design; no empirical validation through real-world BECCS pilot data or quantitative modelling.

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	integrity-preserving governance mechanisms.		Granularity: Coarse; <i>Temporal</i> : In 2023-24 <i>Spatial</i> : Sweden		and threaten environmental integrity. Recommends implementing proportional attribution and adjusting national GHG accounting to prevent double-claims.	uniquely bridging state support mechanisms with voluntary carbon accounting systems.	
Kuhn and Uler (2019)	Identified behavioural motivations influencing individuals' decisions to purchase carbon offsets and to assess how the introduction of offsets affects emission-related behaviour.	Controlled laboratory experiment, behavioural analysis	Primary data from survey and experimental behavioural observations; Granularity: Fine; <i>Temporal</i> : 2012 <i>Spatial</i> : USA	Laboratory experiment conducted at the University of Michigan, USA to understand buying behaviour of Individuals (for carbon offset)	Demand for offsets remains stable at low prices; behaviour is heterogeneous- individuals with higher personal responsibility buy more offsets when facing greater personal damage; offset availability did not reduce actual emissions.	The methodology design, which applies a controlled laboratory experiment to isolate personal responsibility and its effect on voluntary carbon offset purchases.	The subject pool was university students, limiting generalizability beyond that demographic.
Miller et al. (2014)	Explored U.S. family forest landowners' interest, perceptions, and motivations regarding participation in forest carbon offset programs.	Thematic analysis	Primary data captured from focus group participants, survey; Granularity: Fine; <i>Temporal</i> : In 2012 <i>Spatial</i> : Michigan, Minnesota, Wisconsin, USA	Participation of family forest land owners in VCMs	Landowners showed interest primarily when carbon offset programs aligned with personal land stewardship values and provided co-benefits such as habitat, recreation, or tax incentives. Low payment levels, contract duration, and perceived complexity were major deterrents.	The qualitative approach offers in-depth understanding of landowner values and decision-context, revealing motivations that quantitative surveys might miss.	The authors note that focus group participants were a self-selected subgroup responding to the earlier survey, which may bias results toward more engaged or informed landowners.
Park and Oh (2016)	Addressed the lack of scientific data and feasibility assessment for implementing REDD mechanisms in Korea, where deforestation and forest degradation are under-researched.	Baseline approach estimation	Secondary data from govt. reports, previous studies, institutional databases; Granularity: Coarse; <i>Temporal</i> : 2011–2030 <i>Spatial</i> : North Korea	Implementation of REDD mechanism in Democratic People's Republic of Korea (DPRK) to achieve climate goals	REDD mechanisms have significant potential to prevent forest degradation and generate carbon credits, but need careful policy design. The Incentive Accounting Approach was found to yield the highest carbon credits, suggesting its effectiveness under global benchmark settings.	It is one of the first scientific attempts to model REDD feasibility in DPRK using national forest data and multiple baseline approaches.	The revenue estimates do not account for implementation costs, governance risks, transaction expenses, or opportunity costs, hence the economic feasibility remains inconclusive.
Contasti et al. (2023)	Assessed whether carbon credit income can truly offset potential yield losses from adopting climate-smart agriculture practically.	Bayesian decision-analytic framework, structured decision-making approach, scenario analysis	Secondary data sourced from past studies, national reports, and carbon credit registries; Granularity: Coarse; <i>Temporal</i> : Focused on multi-year crop cycles <i>Spatial</i> : Mid-Southern United States	Climate-smart actions (no-till and cover crops) seen as potential options to reduce carbon emissions and promote soil organic carbon storage when linked to VCMs, but requires more scientific examination.	No-till combined with existing cover crop use minimizes yield income loss, and projected carbon credit income can compensate for yield deficits. 2 Helps identify risk thresholds and informs policy design for aligning climate and livelihood goals.	Uses Bayesian decision analysis to integrate agronomic, soil, and economic variables, empowering producers to assess profitability under carbon-financed climate-smart practices.	Relies on aggregated secondary data that may not reflect site-specific conditions.
Aneseyee et al. (2022)	Mapped spatial and temporal variation in ecosystem carbon storage and expressed it	Land use classification, Biomass estimation, Statistical analysis and modelling	Primary data collected through field surveys; Secondary data from laboratory analysis, satellite imagery,	Spatial and temporal variations in carbon storage and valuation in the upper Omo Gibe Basin	Major carbon loss occurred in forests and grazing land due to deforestation and agricultural expansion. Forest land had the highest	Combines field-based carbon estimation with remote sensing and economic modelling, offering policy-relevant	Current carbon pricing approaches are controversial, and valuation methods vary

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	economically to inform sustainability planning.		online databases; Granularity: Fine; <i>Temporal</i> : 1988–2018 (30 years, analysed at four time points) <i>Spatial</i> : Upper Omo Gibe Basin, Ethiopia		total carbon, and cultivated land the lowest.	insights for REDD+ and CDM implementation in Ethiopia.	significantly across regions.
Watson et al. (2013)	Examined how uncertainty in forest carbon stock estimates affects projected emission reductions and economic viability of a REDD project in Ethiopia's Bale Mountains.	Revenue scenario analysis, sensitivity analysis	Primary data collected through field surveys; Secondary data from carbon stock values; Granularity: Fine; <i>Temporal</i> : 20-year project timeframe <i>Spatial</i> : Bale Mountains Forest in Ethiopia	Uncertainty of emission reduction in REDD based forest conservation efforts in Bale Mountains, Ethiopia	IPCC biome-average values underestimate regional carbon stocks by up to 63 % in moist forests and 58 % in dry forests. Using primary data can enhance estimated emission reductions, significantly increasing projected carbon credits and revenues.	Demonstrates how uncertainty in forest carbon stock data translates into large variations in REDD, emphasizing the need for improved data and thus strengthening both the scientific and financial credibility of REDD initiatives.	There can be an opportunity to include implementation costs, governance, or leakage risks in REDD.
Karavai and Hinostroza (2013)	Explored how VCM actors in sub-Saharan Africa conceptualize sustainability, and which sustainability discourses dominate project design and standards.	Qualitative discourse analysis	Primary data collected from interviews, survey responses; Secondary data collected from PDDs, registry database; Granularity: Coarse; <i>Temporal</i> : In February 2009 and June 2011 <i>Spatial</i> : 23 countries in Africa	Assessment of sustainability parameter of seventy-eight VCM projects across twenty-three countries in sub-Saharan Africa	57 % of projects reflected weak ecological modernization (WEM) while only one project followed sustainable development (SD) discourse. Discursive powers skewed toward Northern actors. Sustainability was framed largely around efficiency, market value, and cost-effectiveness.	The study provides a novel discursive framework for analysing sustainability in carbon markets.	The sample might not be representative of all VCM activities in sub-Saharan Africa, as the study only included publicly documented projects.
Mathur et al. (2014)	Addressed the limited understanding of how host communities in developing countries experience and engage with carbon market projects, particularly focusing on the justice implications.	Multi-case study approach, participatory rural appraisal, development of a four-axis analytical framework	Primary data collected through interviews, survey responses; Secondary data from PDDs, registry database; Granularity: Fine; <i>Temporal</i> : Data was collected between 2008 and 2012 across projects initiated in the 2000s <i>Spatial</i> : Sub-Saharan Africa and South/Southeast Asia	VCM projects poses equity and justice dilemmas for communities in Asian and Africa countries	Host communities generally receive limited benefits and lack agency in project governance as procedural representation is weak; power asymmetries favour international and national actors. Capacity-building and third-party facilitation can improve justice outcomes.	The study presents a multi-scalar justice framework and empirical grounding in four distinct case studies across continents.	Enhancement in the number of case studies and assessment of the later stages of the projects can provide longitudinal outcomes of the projects.
Teo (2024)	Examined how different actors within VCM conceptualize the purpose and moral legitimacy of carbon offsetting, especially regarding the tension between emissions reduction and co-benefits.	Inductive thematic analysis	Primary data from semi-structured interviews; Secondary data from public reports, white papers, organisational statements; Granularity: Fine;	Different VCM participants may have different moral understanding possess about carbon offsetting and its goals.	Tensions arise from the need to make emissions and offsets commensurable while acknowledging project-specific complexities. Co-benefits are central to legitimacy for many developers.	Extends moral economy frameworks to VCMs and empirically captures the divergent worldviews of market participants.	The interview-based research approach may be limited by attitudinal fallacy i.e. participants' self-reported justifications may not align with actual behaviour.

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Aryal et al. (2024)	Analysed core emerging challenges that REDD + must address to ensure its effectiveness and legitimacy in forest conservation and carbon mitigation.	Argumentative Policy discussion, critical analysis across REDD + literature	<i>Temporal:</i> In 2023 <i>Spatial:</i> Global Primary data from interviews; Secondary data from past studies, project documents, govt. reports; Granularity: Fine-medium; <i>Temporal:</i> 2023 <i>Spatial:</i> Global (multi-national REDD + contexts)	As REDD remains a globally renowned conservation initiative, its implementation and performance remains debatable amidst the emerging VCM landscape.	Through identification of ten critical questions, and subsequent analysis, the study neither advocates that REDD + would be instrumental in emission reduction nor claims that REDD+ is not important in carbon sequestration. Emphasizes the need for adaptive governance and stronger integration with local communities.	The study digs deep the REDD + associated issues that need to be informed to the wider audience, including decision makers, climate negotiators, scholars, and civil society actors.	There is future scope for recognition of historical entitlements and contributions of indigenous people and local communities in REDD + instruments that is currently not adequate.
Valiergue and Ehrenstein (2023)	Explored how offset suppliers navigate skepticism and moral scrutiny by developing self-regulation, co-benefits framing, and client relationships, which are often absent in technical or econometric assessments of carbon markets.	Qualitative analysis	Primary data from interviews; Secondary data from market reports, press articles, academic literature; Granularity: Fine-medium; <i>Temporal:</i> 2007–2022 <i>Spatial:</i> Global	The credibility of offsetting as a solution to climate change remains contested, as big companies resort to purchasing carbon credits.	Highlights that quality and while corporations increasingly demand higher rigor, standardized protocols are costly and may limit market growth. Emphasizes the trade-off between transparency and scale.	Sheds light on self-regulation in the markets, the production of offsets with co-benefits from feasible projects, and, finally, the relationship between offset sellers and their clients.	Interviews were conducted mainly between 2012 and 2017, so some of the latest market developments are not directly observed but were discussed via secondary sources.
Lee et al. (2015)	Examined gendered equity in access, decision-making, and outcome distribution within the Kenya Agricultural Carbon Project.	Qualitative case analysis using three-tier equity framework	Primary data from interviews, field observations; Secondary data from project documents, previous studies; Granularity: Fine; <i>Temporal:</i> In 2012 <i>Spatial:</i> Multiple communities in Kisumu and Kitale counties in western Kenya Western Kenya	Most carbon projects pay little attention to equity implications for underrepresented community members including women	Women faced barriers to access due to land tenure norms and household decision hierarchies, despite being primary farm labourers while men retained control over major farm decisions and community-level benefit. Nonmonetary co-benefits (e. g., food security, empowerment) were significant, especially for women; carbon payments alone were too small to drive participation.	Provides holistic understanding of carbon projects interaction with social structures through integration of gendered equity framework.	Upstream participation (in project design) was absent for both men and women. Hence, project-wide strategic decisions remained top-down, potentially constraining long-term equitable outcomes.
Wang and Corson (2015)	Examined how clean cookstove carbon projects are constructed, packaged, and promoted as “charismatic” offset tokens, highlighting gender and power dynamics.	Ethnographic case approach, document analysis	Primary data from interviews; Secondary data from PDDs, emission reduction purchase agreements, registry information; Granularity: Fine; <i>Temporal:</i> Preliminary visit in July 2010; main fieldwork in June–August 2011 <i>Spatial:</i> Western Kenya	Role of rural Kenyan women undermined in carbon projects	The study reveals power asymmetries embedded in VCM narratives and credit commodification as cookstove projects rely on women’s labour to generate measurable emission reductions, transforming their domestic work into tradable credits symbolising false empowerment.	A critical examination of how gendered labour and local adoption dynamics are essential yet overlooked in the creation of carbon credits.	The study is based on a single case, and while theoretically rich, generalizability to other contexts requires caution.

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Kim (2022)	Examined whether the presence of female executives and employees influences South Korean firms' decisions to voluntarily disclose carbon emissions information.	Regression analysis	Secondary data sourced from company annual reports, ESG reports, online databases; Granularity: Coarse; Temporal: From 2014 to 2020 Spatial: South Korea	As female representation in the Korean firms increase, it is likely to have an impact on voluntary carbon disclosure by the firms and also their ESG investment decisions	Firms with a higher proportion of female executives and employees were significantly more likely to voluntarily disclose carbon emission information suggesting that gender-diverse governance enhances transparency and climate-related reporting.	Enhances understanding of gender related ESG impact by including both female executives and general female workforce metrics.	Does not address potential omitted variable bias due to unobserved factors that may influence both female representation and disclosure behaviour.
Gay-Antaki (2016)	Examined how carbon offset interventions intersect with gender, land rights, and indigenous practices, rather than focusing only on ecological or economic dimensions of market-based conservation.	Ethnographic case study, document review	Primary data from interviews; Secondary data from PDDs, policy texts, past studies; Granularity: Fine-medium; Temporal: In 2014–2015 Spatial: Focused on multiple community sites in Oaxaca, Mexico	Two Indigenous community sites in Oaxaca, Mexico, one forest-based and one wind-based carbon market project	In both cases, women disproportionately contributed unpaid labour but remained excluded from decision-making exposing how market-based conservation deepens gender inequality under the guise of sustainability.	Applied a feminist political-ecology framework to critically examine carbon forestry projects to highlight gender concerns.	Findings are based on a small number of indigenous communities and a qualitative approach, limiting the generalization of the findings.
Wissner et al. (2022)	Assessed the co-benefits and negative externalities of five VCM project types using standardized impact indicators and expert validation.	Comparative project-level analysis using Sustainable Development (SD) impact scoring framework, expert judgment and review of 4–6 sample projects per type	Secondary data from previous studies, expert judgement, registry documents, project reports; Granularity: Coarse; Temporal: Not specifically defined Spatial: Global	VCM projects (globally) may address impacts of projects beyond climate change mitigation	Cook stove projects scored highest, while landfill gas and renewable energy scored lower in SD performance.	The study provides a comparative evaluation of multiple project types based on a unified SDG-based framework.	Future research addressing concerns like additionally, non-permanence, and quantification of emission reduction needed.
Lehmann (2019)	Analysed how marketing of Gold Standard-certified cookstove projects frames co-benefits, especially for women, through a gender-sensitive, critical political economy lens.	Qualitative content analysis of marketing materials of 22 Gold Standard cookstove projects	Secondary data from Gold standard project documents; Granularity: Fine; Temporal: In 2018 Spatial: Global	VCM projects in the global south claiming improved income and women empowerment	VCM projects may falsely claim improved income and women empowerment, undermining the agency of women	With a strong theoretical grounding in Cultural Political Economy, the study uncovers how carbon projects are publicly framed.	Study primarily based on marketing materials but lacked perspectives from the concerned stakeholders

Source: Authors' compilation from the selected studies for the review

6. Conclusion and future research direction

Based on the investigation of the selected corpus of VCM literature, this study found some meaningful insights that fulfilled the purpose behind structuring and synthesizing the selected sixty-nine studies. The foremost requirement of the VCMs, which stays common throughout the extant literature and across all regions, has been a need for standard policy framework that makes the operations and governance of projects effective and reasonable. The COP 29, which culminated in Baku in 2024, considerably addressed this requirement at the international level. However, much work still needs to be done at the national, regional, and international level.

To conclude the findings based on climatic zone wise review, studies conducted in the Tropical climatic zone reveal mixed outcomes when it comes to returns from the VCM projects. While some studies showed potential for increased returns, several studies showed economic viability of such projects limited by several factors, ranging from context specific factors to the price of carbon credits. Furthermore, the studies emphasize governance issues and transparency in the markets, urging for regulation. The studies specific to the Temperate zone (mostly performed in the Global North) showed that forestry projects could be profitable if sustainable practices were introduced. Additionally, the VCM projects were seen to have a comparatively higher opportunity cost of implementation. In case of studies conducted in the Arid zone, a limited number of studies were conducted broadly covering agroforestry and the agriculture sector, revealing agroforestry projects could be cost-effective.

The studies in the continental zone focussed on examining individual responsibility towards voluntary carbon offsetting. A few studies also used cost-benefit analysis of carbon credit offsets to show that current carbon prices are not likely enough to encourage landowners to participate in the carbon markets. Furthermore, studies belonging to multiple climate zones emphasized the examination of the cost-effectiveness of the VCM projects. The review of the studies, when seen through the lens of geographical location of the studies, revealed a considerable difference in the pattern of the studies belonging to the Global North vis-à-vis Global South. The types of projects studied, their intended research objectives and outcomes, and the location of the studies considerably mirror the non-geographical divide.

The thematic assessment of the studies revealed that VCM projects are found to revolve around optimal land use to determine sustainable, financial, and economic usage because of the limited availability of the asset. Several studies investigated the most beneficial plantation for a specific piece of land and offered mixed opinions based on their results. Thus, while setting up policies for VCM and land use, past empirical evidence related to optimal land use can be studied to inculcate feasible, profitable, and socially constructive alternatives for sustainable development. Sustainable development is represented by rational choices for the betterment of future generations and the significant gain of communities engaged in the process. Community betterment was mentioned as one of the several co-benefits of VCM projects which also became the main highlight for the sale of carbon credits even in the event of missing profitability. However, the integrity of such co-benefits and disbelief in the concept of carbon offsetting was questioned. Hence the issue of poor governance needs to be resolved for the host communities to benefit from climate change actions.

While examining the role of women in VCM projects, it is remarkable to see the stereotypical narratives being woven around carbon credits in the name of “charismatic carbon”. Such narratives are used only to add value to the carbon credits sale but undermine the agency of the women. These observations requisite the need to delve deep into the gendered dimensions of VCM projects, further linking it to broader socio-economic and climate goals.

Though the criticism faced by VCMs received some setbacks from the advancements made in COP 29, grey areas such as the undisclosed locations of the projects in some cases, lack of transparency in operations,

unnamed beneficiaries, gender inequality narratives, and no reporting of the projects still pose numerous problems. Such challenges are serious roadblocks to the VCM financing mechanism and hampers decision-making by stakeholders. Addressing such concerns requisite more empirical studies from different regions and from different perspectives to widen the evidence-base for policy-making. The researchers can undertake case-based investigations (including multi-country cases) by visiting the identifiable project locations. Longitudinal studies that track the outcomes of carbon offset projects over time, including permanence, leakage, and actual versus projected carbon sequestration should be encouraged. Studies focusing on developing approaches to define baseline or improving existing approaches to measure carbon offsetting would be highly instrumental. Additionally, research should focus on measuring co-benefits (like income generation, biodiversity, water access). Studies examining the role of technology such as the use blockchain, remote sensing, GIS, and satellite imagery to assess changes in land use, forest cover, or agricultural practices will help address concerns related to additionality and transparency in VCM projects.

The research in the VCM domain is gaining momentum, as seen in the research timeline of previous studies in VCM with fifteen studies published in 2024. A critical appraisal of the key findings also reveals the potential areas of exploration in this field. To contribute towards the research conducted in this area, an attempt can be made to study ongoing VCM projects in sectors of interest to investigate the financing mechanism of such projects, their profitability, and the amount of carbon being sequestered through them. Furthermore, researchers interested in the policy area can perform a comparative analysis of the carbon markets policy across various nations. Such comparisons are likely to help identify good practices from different markets, which can be tailored to develop markets in nations or regions lagging. Additionally, cost and benefit analysis can be performed for various VCM projects, and bringing in land use perspective could guide better land use decision to meet climate goals. Scholars can also work on identifying possible nexus formations that might be operating in the VCM value chain.

CRedit authorship contribution statement

Ridhi Kochar: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anjala Kalsie:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Nabajyoti Deka:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2025.146261>.

Data availability

Data will be made available on request.

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